Trade Attitudes In The Wild

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Abstract: Scholars are increasingly turning to the role of information to explain why individuals' economic interests often fail to predict their economic policy preferences. But what kinds of economic information are citizens exposed to in the real world? We combine a thirteen-month national panel survey in the United States with individual-level behavioral measures of media consumption, to explore what news about trade Americans are exposed to in a naturalistic setting, and how it shapes their trade preferences. We find that most Americans are exposed to relatively little news about trade, but that the kind of trade news Americans are exposed to in the real world does not magnify the effects of economic interests. Instead, we find some evidence that trade news affects trade preferences through sociotropic rather than pocketbook pathways: Americans become more supportive of trade the more positive stories about trade they see.

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1 Introduction

One of the central principles in political economy is that actors' policy preferences are driven by their economic interests (Rogowski, 1987; Frieden, 1991; Lake, 2009). Policies will be supported by those who stand to gain from them, and opposed by those who stand to lose. One of the central puzzles in contemporary political science, then, is why these theories of economic interests seem to perform poorly at the individual level, especially when it comes to preferences about trade (Herrmann, Tetlock and Diascro, 2001; Mansfield and Mutz, 2009; Naoi and Kume, 2011; Rathbun, 2016; Guisinger, 2017).

These lively debates about the limits of self-interest have led to a new wave of scholarship that seeks to explain *why* economic interests fail to predict economic preferences, frequently pointing to the role of information (Guisinger, 2017; Rho and Tomz, 2017; Schaffer and Spilker, 2019; Jamal and Milner, 2019). According to one strand of arguments, economic interests fail to predict support for trade because ordinary citizens simply don't know any better. Unburdened by information about trade's costs and benefits, and unacquainted with the teachings of Heckscher-Ohlin or the revelations of Ricardo-Viner, ordinary citizens end up structuring their trade preferences through a range of other factors, from demographic characteristics like race and gender (Guisinger, 2017) to psychological dispositions like ethnocentrism or isolationism (Mutz and Kim, 2017). Using carefully controlled experiments, political scientists have found that economic models of self-interest perform better when individuals are taught the precepts of economic theory (Rho and Tomz, 2017; Bearce and Moya, 2020).

In directing us to the role of information, these experiments perform an invaluable service. Yet they leave unexamined an important question: what kinds of information about trade are citizens exposed to in the real world? And, what effect does this information have on how individuals think about trade? In this study, we revisit the relationship between economic interests, psychological dispositions, trade preferences, and individuals' information environments, fielding a seven-wave panel survey on a national sample of Americans over a thirteen month period. Rather than experimentally inject our respondents with information to see what effect it has on their trade preferences, our interest is in studying Americans' information environment about trade in the wild. To that end, each of the respondents in our nationally representative sample consented to installing software on their computers, phones and/or tablets that passively tracked their online activity in real-time during

the fielding of our study. In this manner, we are able to reconstruct each respondent's information environment at the individual level, URL by URL, letting us determine both how much, and what kinds, of news about trade individuals are exposed to in their daily lives over a several month period, and enabling us to test what effect this more naturalistic form of information has on their trade preferences.

The discussion below has three parts. We begin by briefly reviewing debates about the limits of material interests in predicting trade attitudes, showing how some scholars have sought to rescue these models by pointing to the crucial role of information. Yet this approach leaves unanswered questions about how to best study information in a naturalistic setting, and how the type of information subjects are often presented with in IPE experiments relates to the types of information about trade Americans receive in the wild. We then discuss the methods of our study, designed to overcome some of the obstacles that hinder the study of media effects and information exposure more generally, before presenting our results. We show just how far removed most Americans are from information about trade: even though our study was fielded during a trade war in 2018-19 when trade was all over the news, nearly 60% of our respondents never once accessed a news article mentioning trade. Yet we also show that the kind of information about trade the public is likely to receive in a naturalistic setting does not necessarily magnify the effects of economic interests. Instead, we find evidence that news about trade in the media affects trade preferences through sociotropic rather than pocketbook pathways: the more positive stories about trade in general that respondents encounter, the more supportive of trade they become.

2 Material interests and their discontents

The central organizing framework in the study of International Political Economy (IPE) in the United States is the Open Economy Politics (OEP) framework, which holds that where you stand on economic issues depends on where you sit in the global economy (Lake, 2009; Cohen, 2007). Actors have preferences over outcomes, based on their economic interests; these interests are then aggregated through domestic institutions to form national policy, and shape outcomes on the international stage through interactions (Milner, 1997; Lake, 2009). This framework has many virtues, but two are especially notable. First is its generalizability: it applies as much to individuals as to firms or sectors, as much to the movement of goods (Davis, 2003; Milner and Kubota, 2005) or money

(Pandya, 2010) as to the movement of people (Peters, 2017). Second is its specificity: because actors' interests can be derived from the predictions of economic theory, political scientists can motivate their models using theoretical frameworks like Stolper-Samuelson, Ricardo-Viner, or Heckscher–Ohlin, which make determinate predictions about actors' preferences (Rogowski, 1987; Hiscox, 2001). It thereby facilitates the linkage between economics and politics: the former explains what actors want, the latter what they get.

However, the track record of economic interests in predicting what individuals want, especially with regard to trade, is mixed (Kuo and Naoi, 2015). An earlier wave of work, often using education as a proxy for respondents' skill level, found a correlation between economic interests and trade attitudes, showing that the individuals who were the most supportive of free trade tended to be the ones who had the most to gain from it (Scheve and Slaughter, 2001; O'Rourke et al., 2001; Mayda and Rodrik, 2005). Yet as Hainmueller and Hiscox (2006) note, education is a somewhat blunt measure of economic interests, capturing a variety of other mechanisms, ranging from knowledge, to psychological traits, to cultural values. And, a subsequent wave of work has generally been much more pessimistic. Mansfield and Mutz (2009) find that trade attitudes are sociotropic rather than egocentric; Herrmann, Tetlock and Diascro (2001) and Rathbun (2016) show that support for free trade has ideological roots rather than traditional economic ones; other scholars similarly point to the important role of social trust (Kaltenthaler and Miller, 2013), beliefs about the government's role in the economy (Wu, 2019), cognitive style and psychological needs (Johnston, 2013), prejudice and cultural factors (Margalit, 2012; Sabet, 2014), and so on. As Hafner-Burton et al. (2017, S2) note, "after more than a decade of careful empirical research, there is little evidence that voters actually define their interests in these rational, materialist ways."

The weakness of economic interests in predicting trade attitudes is consistent with the political behavior literature more broadly, which apart from a few notable examples (such as smokers opposing smoking bans - Citrin et al., 1997) has traditionally found little evidence in favor of personal pocketbook-based explanations of political attitudes (Feldman, 1982; Kinder, Adams and Gronke, 1989; Mansbridge, 1990). And, these questions have also taken on additional resonance given recent debates about whether to attribute the rise of populism and backlash against globalization to economic interests or cultural variables (Norris and Inglehart, 2019; Frieden, 2022).

Defenders of OEP have responded to these critiques in at least three different ways. The first is methodological: if studies employ high-quality and extensive measures of psychological traits, but

relatively low-quality and limited measures of economic interests, it perhaps should not be surprising that the former will outperform the latter (Fordham and Kleinberg, 2012; Kuo and Naoi, 2015; Owen and Walter, 2017). The second involves showing that cultural and psychological variables can have economic causes, echoing earlier Marxist arguments about the relationship between material base and ideational superstructure. Thus, for example, a series of articles have attributed the rise of economic nationalism, authoritarianism, and support for economic disintegration to import competition from China (Autor, Dorn and Hanson, 2013; Ballard-Rosa et al., 2021; Colantone and Stanig, 2018). Rather than viewing economic and cultural variables as theoretical rivals, then, this literature encourages us to think of them as complements in a broader theoretical chain.

The third, and of particular importance here, points to the role of information. While foreign policy issues in general are typically understood to be further removed from most citizens' daily lives (Rosenau, 1965), this is especially true for foreign economic policy. Trade is a "hard" rather than an "easy" issue (Carmines and Stimson, 1980), a subject about which, at least a decade and a half ago, few voters thought about (Guisinger, 2009), and even fewer understood (Hiscox, 2006, 758), as the public consistently displays high levels of "economic ignorance" (Caplan, 2011). A string of surveys in the mid-to-late twentieth century found that only 18% of Americans could define monetary policy, 39% could define free trade, and 51% could describe the effects of tariffs (Delli Carpini and Keeter, 1996, 70-71). As Guisinger (2017, 76) notes, "information is the foundation of policy preferences." If the public isn't necessarily aware of what free trade is, they presumably are hazy on the finer points of "new" new trade theory as well, and may be less equipped to connect their personal economic interests with complex economic policies.

As a result, a wave of important studies has appeared in the past several years seeking to examine what "fully informed" (Althaus, 1998) preferences about foreign economic policy might look like, frequently using experimental methods. Guisinger (2017) finds that support for trade increases once you correct misperceptions about who America's most important trade partner is, Bearce and Tuxhorn (2017) find that monetary policy preferences become more aligned with respondents' material interests when respondents are given more contextual information explaining how monetary policy works, Rho and Tomz (2017) find that respondents' trade preferences tend to align with Stolper-Samuelson when they are taught its precepts, and Jamal and Milner (2019) find similar effects with Ricardo-Viner in Tunisia. Herrmann, Tetlock and Diascro (2001); Hiscox (2006); Ardanaz, Murillo and Pinto (2013) and Guisinger (2017) also experimentally manipulate respondents' information

environments in different ways, finding that respondents presented with positive frames about trade typically espouse more pro-trade preferences than those presented with negative frames, although there is also important individual-level heterogeneity, including along race and gender.

In providing causal evidence that respondents' information environments shape their economic preferences, this wave of scholarship makes a crucially important contribution. It also highlights a crucial limitation of OEP frameworks that assume individuals think like complicated economic theoretical frameworks to which they may never have been directly exposed; indeed, if respondents in the control conditions in these experiments were already familiar with Stolper-Samuelson or Ricardo-Viner, presumably these studies' treatments would be introducing no new information and should have little impact. Yet these studies also raise a broader question. These experiments tell us how respondents would respond to treatments *if* they were to receive them (Althaus et al., 2011). In the real world, however, voters are rarely given crash courses in economic theory; elites may strategically employ frames, but in a naturalistic information environment these frames often cancel one another out (Druckman, 2004). If it is true that, as Mansfield and Mutz (2009, 453) argue, "a better understanding of trade preferences among the mass public requires that we account for the broader information environment", it is valuable for us to determine what respondents' information environments look like in a more naturalistic setting. What information about trade are ordinary citizens exposed to in the real world?

3 Information exposure in the wild

Political scientists interested in studying information exposure typically have three approaches to choose from. First, we can use group membership as a proxy for information exposure. In a clever series of studies, Ahlquist, Clayton and Levi (2014) and Kim and Margalit (2017) show the important role of labor unions in shaping their members' attitudes towards free trade, such that as union leaders changed positions on trade, union members did too. Second, we can study the supply side of the information environment, usually through painstaking content analyses of news coverage (Hayes and Guardino, 2011; Baum, 2002; Guisinger, 2017), which tell us both what stories the media covers, and what frames the media uses in its coverage. Third, we can study the demand side of the information environment, either studying respondents' media consumption directly using self-report items, or employing related variables like education or political interest (Zaller, 1992;

Gadarian, 2010; Berinsky, 2009)

Yet there are a number of challenges here. On the supply side, the challenge is one of "assumed transmission" (Althaus et al., 2011): media coverage is necessary, but not sufficient, to assume that citizens are receiving the treatment, especially if many individuals pay relatively little attention to foreign policy issues. And, because of the amount of time involved in coding media coverage, many supply-side analyses understandably typically focus on a small number of prominent outlets, as in front page stories of the *New York Times*, or transcripts of nightly national newscasts on the "big three" networks. In an era in which nightly newscasts are seeing declining viewership rates and Americans are increasingly getting their news online through social media, we may wish to cast a wider net (Boukes, 2019).

On the demand side, the challenge is that media consumption is extraordinarily difficult to measure. Just as participants in nutrition studies tend to overreport how many vegetables they eat, survey respondents in political science tend to significantly overreport how much news they consume. Prior (2009), for example, shows that whereas Nielsen data suggest between 30-35 million people watched the nightly news on an average weekday in the year 2000, ANES self-report data from that year would suggest that 85-110 million people watched a night. It is perhaps for these reasons that observational studies traditionally found minimal evidence of media effects, a finding that Bartels (1993, 267) colorfully characterized as "one of the most notable embarrassments of modern social science."

In this study, we take a different approach, fielding a panel survey on a nationally representative sample of Americans, and integrating it with web tracking data that gives us behaviorally validated individual-level measures of their online media consumption, thereby avoiding the challenges both of assumed transmission and self-report biases (Clemm von Hohenberg et al., 2024). The closest existing work to ours in an IR context is Pelc (2013), who uses monthly state-level Google search data as a behavioral measure of information-seeking to test theoretical assumptions in IPE; we obtain a similarly behavioral measure of information consumption to address questions in IPE, but at the individual-level, and in real-time. Our interests here are twofold: how much news about trade are ordinary citizens exposed to? And, how does it shape their trade preferences?

3.1 Information exposure and trade preferences

Most contemporary models of the effects of information exposure on citizens' policy preferences are based on some version of the expectancy value model of political attitudes (Ajzen and Fishbein, 1980), which represent attitudes towards a given policy as a weighted sum of evaluative beliefs $\sum(v_i \times w_i)$, where v_i represents the value the citizen assigns the policy on a given dimension (i), and w_i represents the weight the citizen places on it. Thus, persuasion involves changing what people think by providing information that alters evaluative belief v_i, whereas framing or priming involve changing w_i, how much people emphasize their existing beliefs about that particular dimension relative to others in their overall evaluation of the policy (Iyengar and Kinder, 1987; Nelson, Clawson and Oxley, 1997; Chong and Druckman, 2007). Even without invoking expectancy value models, all information experiments in IPE implicitly target one of these two variables, either seeking to change respondents' beliefs about a given policy issue (e.g. Rho and Tomz, 2017; Jamal and Milner, 2019), or to recalibrate how much respondents weight one consideration over others (e.g. Hiscox, 2006; Guisinger, 2017).[7]

In the context of preferences towards economic issues like trade, there are two ideal typical dimensions through which information can affect trade preferences, which following the economic voting literature (Kinder and Kiewiet, 1981; Feldman, 1982), we refer to as *pocketbook* and *sociotropic* considerations, respectively.[8] According to theories of economic attitudes that emphasize pocketbook considerations, individuals can be aware of their personal economic circumstances, but still be unable to attribute them to national-level economic policies; as Guisinger (2017, 38) shows, most Americans indicate that trade either has no effect on their own employment, or admit that they don't know what trade's personal effects are. Providing information about the distributional consequences of trade can therefore help people better connect their personal economic interests to their trade preferences, and thus change their evaluation of trade's pocketbook effects. This is one of the logics of trade preferences explored in Rho and Tomz (2017), who find that experimentally presenting respondents with information about trade's winners and losers enables respondents to express trade preferences more in line with Stolper-Samuelson, or Jamal and Milner (2019) and Schaffer and Spilker (2019), who find the same with regard to Ricardo-Viner. Bearce and Hart (2017) obtain a similar finding

[7] Of course, it is often difficult to distinguish between framing and information provision in practice (Naoi and Kume, 2011), and many framing experiments in IPE also provide respondents with information (e.g. Herrmann, Tetlock and Diascro, 2001; Hiscox, 2006; Jensen and Shin, 2014; Guisinger, 2017; Chilton, Milner and Tingley, 2020).

[8] As we discuss below, scholars sometimes also examine community-level economic interests (e.g. Guisinger 2017; Autor et al. 2020), which we discuss and investigate empirically in Appendix §B.4.

in regards to monetary policy: voters are more likely to express egocentric policy preferences when they are presented with contextual information helping them link macroeconomic policies to their personal circumstances.

The second dimension focuses not on the effects of trade on individuals' own economic circumstances, but rather, whether trade simply has positive effects in the aggregate, thereby benefiting the country as a whole. Economists often argue that the popularity of protectionism in the United States and other western democracies is due to voters' ignorance (Caplan, 2011), a claim echoed by political scientists as well (e.g. Althaus, 1998). Hainmueller and Hiscox (2006, 472) argue that college-educated individuals are more supportive of free trade not because of their economic interests, but because they "are likely to be far more informed than others about the aggregate efficiency gains associated with expanded trade", such that exposure to theories of comparative advantage change individuals' evaluative beliefs about trade's benefits for the country as a whole. Burgoon and Hiscox (2004) similarly argue that women are more protectionist than men because they are less likely to be exposed to economic ideas in college. While operating out of a framing rather than a persuasion tradition, Guisinger (2017) experimentally exposes respondents to positive arguments about trade's effects on the United States, and finds that individuals who receive positive arguments about trade are, on average, significantly more supportive of free trade than individuals who don't. Similarly, Hiscox (2006) experimentally shows that individuals presented with pro-trade arguments are more supportive of free trade than individuals presented with anti-trade arguments.

If we move out of the experimental laboratory and into the wild, we argue that information conveyed by the media is more likely to shape trade preferences through sociotropic dimensions rather than pocketbook ones. The reason is less the limited power of self-interest on policy preferences, but rather, because the media is unlikely to provide information that can help individuals learn about their own economic interests with respect to trade. This is certainly the case when news articles cover the macro-level effects of trade or trade protectionism (e.g. "global growth is forecast to slow as a result of the trade war"[9]), but even granular reporting on how trade impacts particular entities or people teaches uncertain lessons about self-interest. Take, for example, a local newspaper profile of shiitake mushroom farmer Gary Schroeder, who says he had to cut staff and file for bankruptcy after getting "clobbered by imports from both Canada and China" (Wood, 2019).[10] Unless the reader

[9]From https://www.theguardian.com/business/2018/dec/11/imf-financial-crisis-david-lipton.

[10]Indeed, we find that the articles in our sample vary widely in the level of granularity of the information they report about trade. In a hand-coding of 267 randomly-selected articles from our sample that mention "trade", "tariff",

is also a mushroom farmer from Southeastern Pennsylvania, it is less clear what one can learn from this article about one's pocketbook interests regarding trade. At best, the negative valence of the article may leave the reader with an overall impression that trade is hurting hardworking Americans – in other words, a sociotropic effect. Thus, the effects of information exposure in a naturalistic setting are more likely to resemble those in framing experiments (where respondents are presented with information about trade in the aggregate that varies in valence) than those in information provision experiments (where respondents are presented with information about trade's effects on themselves personally). Outside of the laboratory, information can neither be tailored to the economic circumstances of each reader, nor does it typically contain the precepts of economic theory that would help readers work out their economic interests. Indeed, one of the important virtues of experiments is that they allow us to estimate a counterfactual less frequently observed in the real world, where only a third of Americans over the age of 25 have a college degree, most of whom graduate without taking a class in economics.[11]

Nonetheless, each pathway suggests a different set of testable observable implications. If the effect of information about trade in the media primarily operates by changing beliefs about the pocketbook effects of trade, information exposure should magnify the effects of economic interests: individuals exposed to more news about trade should be more likely to express trade preferences consistent with how much they stand to gain or lose from trade. Individuals with high skill levels, employed in industries dependent upon exports, or in jobs with low levels of routinization should become more pro-trade when exposed to more information about trade, while individuals with low skill levels, in industries facing high levels of import competition, or in jobs likely to be offshored should become less supportive of trade when exposed to more information. In contrast, if the effect of information about trade in the media primarily affects trade preferences by changing evaluative beliefs about the sociotropic effects of trade, trade preferences will be associated with information exposure: the more positive stories about trade respondents are exposed to, the more supportive of free trade they'll be, whereas the more negative stories about trade respondents are exposed to, the less supportive of free trade they'll be.

H1 (Pocketbook mechanism): Exposure to trade news moderates the effect of economic

or "tariffs", we find that 38 of the articles discuss the overall impact of trade or trade policies on the U.S. or global economy, 50 discuss the impact on particular sectors/industries, 25 on particular groups (e.g. farmers or workers), 23 on specific companies, and 15 on specific individuals like Gary Schroeder. The categories are not mutually exclusive.

[11] See https://www.census.gov/data/tables/2018/demo/education-attainment/cps-detailed-tables.html, Siegfried (2000).

interests on trade preferences. Respondents exposed to more news about trade will display stronger relationships between their economic interests and their trade preferences.

H2 (Sociotropic mechanism): Exposure to positive (negative) trade news increases (decreases) support for trade.

4 Methods

To analyze the relationship between Americans' information environments and their trade attitudes, we fielded a seven-wave panel survey on YouGov, each on a nationally representative sample of American adults over a thirteen month period.[12] The first wave was fielded beginning in the spring of 2018, from April 23 - July 22 ($N = 3224$), the second from August 28 - September 10 ($N = 1339$), the third from October 5-October 29 ($N = 1180$), the fourth from October 30 - November 6 ($N = 1289$), the fifth from December 20 - January 7, 2019 ($N = 1300$), the sixth from January 24 - February 5, 2019 ($N = 1324$), and the seventh from April 1-9 2019 ($N = 1232$).[13] Our data were thus collected during the height of the 2018-19 US-China trade war, which while not yielding average tariff rates as high as those produced by the new trade war the United States is currently in, were nonetheless unprecedented in the past half century (Flaaen and Pierce, 2025), producing substantial economic effects and attracting extensive media attention.

To obtain behavioral measures of media exposure, we integrate the survey data above with web tracking data (that is, all URLs respondents visit, rather than just domains) passively collected from participants' computers, tablets, and mobile devices through YouGov Pulse. All participants provided consent before installing passive metering software by Reality Mine, which tracks their complete web activity in real-time (apart from passwords and financial transactions, which are excluded from the data).[14] As we show in Appendix §A, there is little evidence that respondents who consented to web tracking systematically differ from the population of American adults as a whole in terms of basic demographics, although as one might imagine, they appear to care slightly less about their data being collected on the internet than does the general population as a whole.

[12] See Appendix §A for a discussion of YouGov's sampling and weighting procedure.

[13] Between waves 3-4 of a survey, an experimental intervention was embedded to study the causal effects of partisan media exposure on political attitudes; it is the subject of a different paper. The large sample size for the first wave is due to a botched roll-out; the effective sample size for that wave is in line with those reported for subsequent waves.

[14] For other recent political science work using similar methods, see Guess (2021); Stier et al. (2022); Munzert et al. (2024).

4.1 Dependent variables

The study has two measures of trade attitudes, based on question wording from Mansfield and Mutz (2009) and Pew Research Center. The first, ENCOURAGE INTERNATIONAL TRADE, asks respondents the extent to which they agree or disagree with the statement that "The United States government should try to encourage international trade with other countries"; the second, FREE TRADE, asks respondents how much they agree with the statement that "Free trade agreements like the North American Free Trade Agreement (NAFTA) have helped the U.S. economy".[15] The two items are moderately intercorrelated ($r = 0.38 - 0.45$, depending on the wave), but because each measures a slightly different construct, we analyze them separately.

4.2 Independent variables

4.2.1 Economic interests

Since one of the critiques of research on trade attitudes has been overly blunt and imprecise measures of economic interests (Fordham and Kleinberg, 2012; Kuo and Naoi, 2015; Owen and Walter, 2017), we sought to obtain granular measures of economic interests consistent with best practices. In a series of open-ended responses similar to those from the General Social Survey (GSS), we asked survey respondents to describe, in their own words, what they did for their current job (or, if not currently employed, their most recent job instead), what their duties consisted of, what their employer is called, and what the industry they work in does. On the basis of these open-ended responses, we assigned an industry code using the 2017 North American Industry Classification System (NAICS) and an occupation code using the 2018 Standard Occupational Classification (SOC) system.[16] Both are coded to the maximum six digits of specificity, or up to as many digits as we felt the level of detail provided in the respondent's description allowed. For each observation, two independent coders assigned industry and occupation codes, and a third coder adjudicated disagreements.[17] We then used these industry and occupation codes to calculate a variety of measures of economic interests

[15] We include mention of NAFTA here to avoid concerns that Americans may not necessarily know what free trade is without providing a prominent example.

[16] For information about the 2017 NAICS coding system, see https://www.census.gov/eos/www/naics/index.html; for the 2018 SOC coding system: https://www.bls.gov/soc/2018/soc_2018_manual.pdf

[17] Respondents not currently working (retired, studying, taking care of home or family, unemployed, temporarily laid off, permanently disabled, or other) received codes according to their *last* job, or received no codes if they did not provide any relevant information. Respondents received codes only for paid work; volunteering or pro-bono work was not coded. Self-employed respondents or respondents working as independent contractors received occupation codes according to the substance of their work rather than as "Chief Executives."

from the existing literature.

Average wage by industry-occupation: Following Mansfield and Mutz (2009, 2013), and Mansfield, Mutz and Brackbill (2019), we calculate average wage by industry-occupation as a proxy measure of respondents' skill level. We use 2017 annual wage data from the U.S. Bureau of Labor Statistics, which is disaggregated by occupation and then by industry within each occupation.[18]

One-year change in average wage by industry-occupation: We calculate the percentage change in average wage by industry-occupation from 2016 to 2017 using annual wage data from the U.S. Bureau of Labor Statistics. We use only wave 1 industry and occupation codings, since the data comes from before the survey period.[19]

Import/export orientation: Following Mansfield and Mutz (2009) and Mansfield and Mutz (2013), we calculate import and export orientation for each NAICS industry. Since trade in goods and trade in services are measured differently, we create separate measures of import and export orientation for goods industries and service industries, though we also create a third variable that merges both measures. We define industry i's import orientation as M_i/Y_i and export orientation as X_i/Y_i, where M_i and X_i respectively represent industry i's annual general imports and annual total exports if i is a goods industry, or imports in services and exports in services if i is a service industry, and Y_i is i's gross annual output. For trade in goods, we use 2017 import and export data by industry from the U.S. International Trade Commission, and for trade in services we use 2017 import and export data from the Bureau of Economic Analysis. We use 2017 gross annual output data from the Bureau of Economic Analysis for all industries.[20]

Import dominance: Based on Mansfield, Mutz and Brackbill (2019), we include a dichotomous variable that takes value 1 if the industry's import orientation is greater than its export orientation and 0 otherwise. We calculate separate measures of import dominance in goods and services

[18] See https://www.bls.gov/oes/special.requests/oesm17in4.zip. We used a weighted iterative matching procedure to obtain the most granular industry-occupation codes possible; we assume correspondence between BLS industry code "999" for federal, state, and local government and U.S. Census Bureau industry code "92" for public administration, but do not assume correspondence at further levels of disaggregation within those industries. For observations missing occupation codes, we take the average annual wage for the NAICS industry code across all occupations.

[19] See https://www.bls.gov/oes/special.requests/oesm16in4.zip. 2016 wage estimates are calculated for each industry according to the same procedure as described above for 2017 wage estimates.

[20] See https://dataweb.usitc.gov/trade/search, https://apps.bea.gov/iTable/iTable.cfm?reqid=62&step=9&isuri=1&6210=4, and https://apps.bea.gov/iTable/iTable.cfm?ReqID=51&step=1. Granularity of this measure for goods industries is limited to the level of detail available in BEA gross annual output data, which varies widely by industry category. For service industries, granularity is limited to three NAICS digits in accordance with the availability of trade in services data. In keeping with Mansfield and Mutz's procedure, we take the natural logarithm of each import and export orientation calculation to mitigate the skewness of its distribution, arbitrarily adding 0.01 to each ratio so that the logarithm is defined for non-traded industries (i.e. industries with 0 imports or exports in 2017). Following the existing literature, we assume that all NAICS industry codes omitted from USITC import and export data and from BEA trade in services data are considered non-tradable.

industries as well as a combined variable.

Offshorability: Following Kaihovaara and Im (2020); Owen (2017); Owen and Johnston (2017), and Mansfield and Mutz (2013), we measure the offshorability of each SOC occupation using Blinder's (2009) offshorability index.[21] Following Owen and Johnston (2017), we create a dichotomous offshorability variable that takes a value of 0 if the occupation has an offshorability measure of 50 or below, or 1 if the occupation has an offshorability measure of 51-100.[22]

Routine task intensity: Following Kaihovaara and Im (2020) and Owen and Walter (2017), we create a measure of routine task intensity (RTI) for each SOC occupation code, replicating Acemoglu and Autor (2011)'s procedure using updated source data. The RTI measure compiles task-level data for each SOC occupation from O*NET to create a measure of the proportion of tasks in an occupation that are routine.[23] The O*NET task measures (current as of January 2019) are aggregated into six composite task measures, which are each standardized by standard deviation and centered at the mean; both parameters are weighted by total employment in each SOC occupation category as calculated via 2017 BLS data.[24] We then subtract the standardized composite measures by their minimum values to re-scale the measures to take only positive values, and and then combine them into measures of routineness, abstractness, and manualness. We calculate RTI using the formula $ln(Routineness) - ln(Abstractness) - ln(Manualness)$.

Union membership: We match each respondent with estimates of the union membership rate in his or her occupation and industry. We use the 2018 edition of Hirsch and Macpherson's (2003) database on union membership by industry and occupation, which is compiled from Current Population Survey microdata.[25]

Two additional close-ended measures, Unemployed and Labor Force, record whether the respondent is unemployed and participating in the labor force, respectively. Unemployed is a dichotomous variable indicating whether the respondent identifies as unemployed or temporarily laid off. Labor Force is a dichotomous variable indicating whether the respondent is participating in the labor force, coded 1 if the respondent identifies as working full or part-time, and 0 otherwise.

[21] Since Blinder uses SOC codes from 2000 to construct his index, we crosswalk his codes to our 2018 SOC codes. Blinder defines occupations with an offshorability score below 25 (on a scale from 0 to 100) as unambiguously "non-offshorable" and does not specify numerical scores for them; we thus conservatively assign all non-offshorable occupations in our sample a score of 24, so as to fall within the lowest category.

[22] For the complete index see the appendix of Blinder (2009).

[23] See https://www.onetcenter.org/database.html#individual-files). For a full description of the O*NET data used to calculate RTI, see Acemoglu and Autor (2011), p. 1163.

[24] See https://www.bls.gov/oes/special.requests/oesm17in4.zip.

[25] See http://unionstats.gsu.edu/Ind_U_2018.xlsx and http://unionstats.gsu.edu/Occ_U_2018.xlsx. We used a weighted iterative matching procedure to obtain the most granular union membership estimates possible.

Finally, while our focus in this article is on individual-level measures of economic interests, as a robustness check we also obtain a number of community-level measures of economic interests, which we describe in detail in Appendix §B.4.

4.3 Dispositional and attitudinal measures

Since trade attitudes have been found to have a number of ideological correlates (Kaltenthaler and Miller, 2013; Rathbun, 2016) we also include a number of measures of respondents' political and foreign policy orientations. Mansfield and Mutz (2009) find that isolationism is an important predictor of support for trade, but since isolationism is usually understood as being one of three different dimensions of foreign policy attitudes (Holsti, 2004; Kertzer et al., 2014; Rathbun et al., 2016), all of which are intercorrelated but capture slightly different phenomena, we obtain measures of all three here, using standard Likert items from the existing literature. For Isolationism, respondents indicated the extent to which they agreed with the statements "The U.S. should mind its own business internationally and let other countries get along the best they can on their own." and "We should not think so much in international terms but concentrate more on our own national problems." For Cooperative Internationalism, or CI, respondents indicated the extent to which they agreed with "The United States needs to cooperate more with the United Nations", and "It is essential for the United States to work with other nations to solve problems such as overpopulation, hunger, and pollution." For Militant Internationalism, or MI, respondents were administered the items "Going to war is unfortunate but sometimes the only solution to international problems", and "The best way to ensure world peace is through American military strength."

Because of the relationship between trade attitudes and intergroup relations (Mansfield and Mutz, 2013; Guisinger, 2017), we obtain two different types of measures of ingroup identification. IR scholars who have studied ingroup identification tend to distinguish between national attachment on the one hand, and national chauvinism on the other (e.g. Herrmann, Isernia and Segatti, 2009), the former implicating the extent to which your nation forms a crucial part of your identity, and the latter examining the tendency to view your nation as superior to others, which Mansfield and Mutz (2009) capture in their emphasis on ethnocentrism versus cosmopolitanism. For National attachment, we borrow two items from Herrmann, Isernia and Segatti (2009): "When someone says something bad about American people, how strongly do you feel it is as if they said something bad about you?", and "How much do you feel that what happens to America in general will be your

fate?", as we do for NATIONAL CHAUVINISM: "How superior is the United States compared to other nations?" and (reverse-coded) "How many things about America make you feel ashamed?". We also include a number of different measures of political knowledge (Delli Carpini and Keeter, 1996), and basic demographic characteristics. Importantly, although our research design is ultimately non-experimental, the panel structure of the survey lets us address some of the questions about reverse causality raised by some of the trade literature (e.g. Fordham and Kleinberg, 2012): all of the dispositional and attitudinal measures discussed above were administered on wave 1 of the survey, whereas the dependent variables for most of the analyses were administered in subsequent waves.

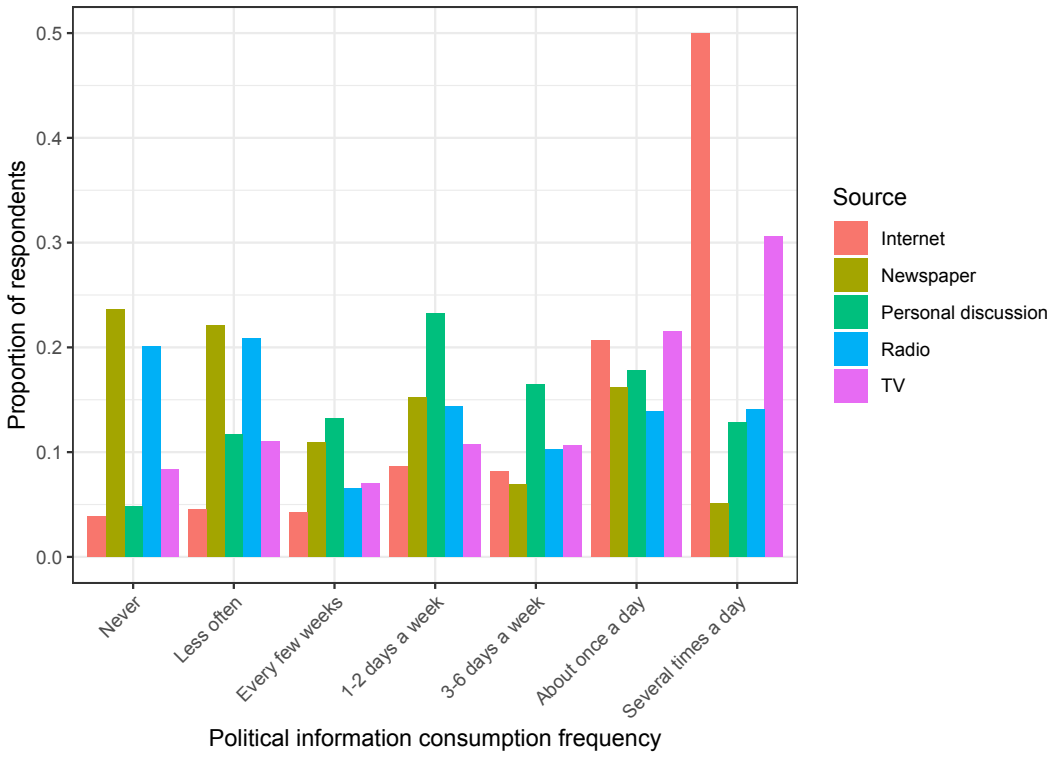
4.4 Behavioral measures

While the measures of economic interests and individual differences discussed above represent best practices in the existing IPE literature, we also innovate by obtaining behavioral measures of media consumption, combining the survey data above with web tracking data passively collected (with respondents' consent) from their computers, tablets, and mobile devices. The internet is not the only means through which individuals can consume political news, but in the twenty-first century it is arguably one of the most important. Figure 1 shows that half of our respondents report consuming political news online several times a day, and 71% of respondents report doing so at least once a day, making it by far the most popular self-reported source of political news in our nationally representative sample, far more than through print media, radio, television, or in-person discussion. Our behavioral approach has important ethical considerations, which we discuss in Appendix §A.2.

As is often the case with "Big Data", the challenge in using behavioral measures of online media consumption is both the volume of information respondents accessed (over the course of the survey, our respondents visited 12,102,415 URLs), and its sheer variety.[26] To obtain tractable measures of respondents' news environments, we restrict our analysis to an updated list of domains identified as "hard news" by Bakshy, Messing and Adamic (2015), who train a classifier on a comprehensive dataset of links shared on Facebook to identify the 500 most shared news sites on the social network in the United States. Restricting the sample in this way not only makes the analysis that follows more tractable, but also provides us with estimates of each domain's ideological alignment based on the ideological leanings of the Facebook users sharing each site. Because we have URL-level data

[26] This figure excludes sequentially duplicated URLs: we employ a de-duplication strategy, where duplicate visits to the same URL in sequence are removed, to avoid overcounting when respondents reloaded their browser.

Figure 1: 71% of respondents report accessing political news online at least once a day



rather than merely domain-level data, we are able to exclude sections of these websites that do not pertain to hard news (e.g. the crossword sections of news sites, the non-news sections of domains like Yahoo and AOL that provide both news and also web-searching and email services, and so on).

The web tracking data are used to produce two sets of behavioral measures of media consumption. The first set focuses on exposure to all hard news, regardless of content. News Count records the total number of hard news URLs visited by each respondent, while Conservative News Count and Liberal News Count record the total number of hard news URLs visited by each respondent from news sites that the classifier in Bakshy, Messing and Adamic (2015) determines to have conservative and liberal ideological slants, respectively. The second set focuses on our chief quantity of interest: news about trade. Our baseline measure of trade news utilizes a simple dictionary-based approach. After removing visits to news sites' homepages to avoid data vintage errors, we scrape the content of each of the remaining 142,841 unique hard news URLs, and use a dictionary-based approach to identify which of these articles discuss trade.[27] Trade News Count merges this data at the respondent-level to record each respondent's total number of visits to articles that mention trade. In addition to this dictionary-based measure of trade news, we also employ two more elaborate measures. First, we use topic modeling and sentiment analysis to obtain more specific measures of the content of these articles; we discuss the setup of the model and the content of the topics it recovers later in the paper. Additionally, as a robustness check, we also build a supervised machine learning classifier trained on news articles that feature trade-related topic tags published by the news outlets themselves, which obtains strikingly similar results; we discuss this approach in detail in Appendix §B.5.

A series of caveats are worth emphasizing here. First, our respondents indicate that the internet is the most important way they receive political news, making studying individuals' online information environments a highly important task. Yet the internet is obviously not the only way individuals can receive information.[28] Second, because we restrict our analysis in this paper to the the 500 most popular hard news sites, we cover a wider range of news sources than is typically used in this type of research (e.g. news sites in the database include large outlets like CNN and the New

[27] We successfully scraped 92.8% of the unique URLs (the remaining 7.2% either yielded errors, or redirected towards each site's homepage. After an initial pilot with a more expansive set of keywords, we settled on a more parsimonious dictionary-based approach that retained any article that included the words "trade", "tariff" or 'tariffs" in the article text; articles that included other trade-related words (NAFTA, etc.) all included one of these three.

[28] In Appendix §B.3 we show that respondents who indicate they're less likely to consume news online also indicate they're less likely to be consuming news offline as well. Our empirical strategy thus rests on the – in our view, plausible – assumption that individuals who aren't exposed to trade news online are also less likely to be exposed to trade news through other means.

York Times, but also include a broad array of smaller outlets, ranging from Electronic Intifada to Reagan Coalition, The Grio to The Federalist). However, these are obviously not the only political news sources available online. Respondents can also receive political information through soft rather than hard news (Baum, 2002), or by passively scanning through their social media feeds (Anspach, Jennings and Arceneaux, 2019). Thus, while the analysis that follows reconstructs individuals' information environments to an extent not usually attempted in the study of international relations, it is inevitably a proxy measure rather than a complete reconstruction. Finally, the mobile data in the analysis below consists only of URLs visited through web browsers rather than through apps; we analyze app data, comparing it with browser consumption, in Appendix §B.3.

5 Results

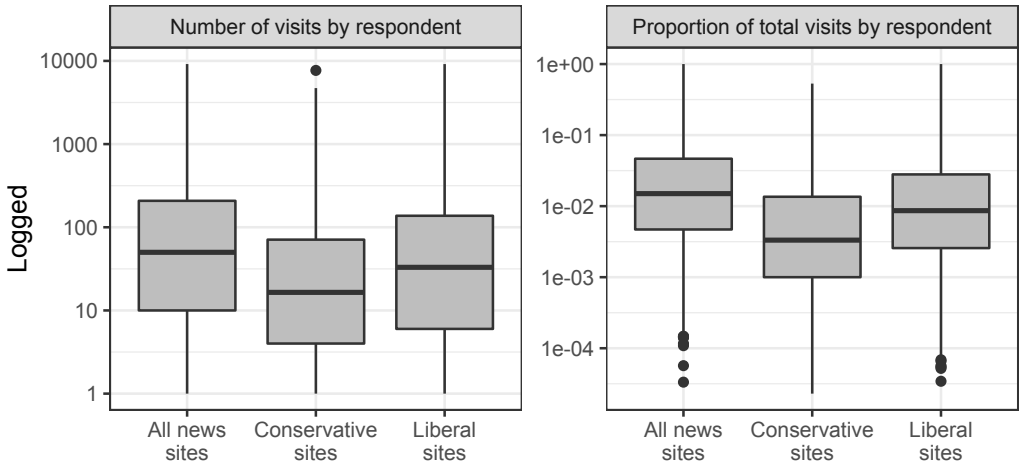
We structure the analysis below in three parts. First, we use our behavioral data to examine how much information about trade respondents are exposed to through online news sources. We show that even during a period when trade was highly salient in the news, most Americans' exposure to news about trade is extremely low, except for the most politically engaged. Second, we test whether exposure to trade news shapes trade preferences through the pocketbook dimension, estimating a series of regression models to explore whether exposure to trade news magnifies the effect of economic interests. We find little evidence in support of this mechanism. Third, we test whether exposure to trade news shapes trade preferences through the sociotropic dimension, estimating a set of fixed effect models estimating the effect of exposure to trade news on within-subject variation in trade preferences, which also provides additional traction on questions of causal identification. We find some evidence in favor of this sociotropic mechanism: respondents who are exposed to more positive news about trade tend to display more positive trade preferences.

5.1 Reconstructing Americans' information environments

To address how much information about trade Americans are exposed to in the wild, we begin by presenting basic descriptives about our behavioral measures of media exposure. Figure 2 presents a series of boxplots depicting the total number of visits each respondent made to news sites (the left-hand panel), and the ratio of each respondent's visits to news sites versus their total online activity (the right-hand panel), based on the Bakshy, Messing and Adamic (2015) domain list described

above. The plots show that although half of Americans in our survey reported consuming news online several times a day, and 71% of respondents reported consuming news online at least once a day, our behavioral measure suggests actual news consumption may be lower than respondents claimed: according to our measure, the median respondent visited only 1.3 URLs from news sites per day. To put this in context, the right-hand panel of Figure 2 shows that although a small minority of Americans are news junkies, most are not: for 26% of respondents, news URLs make up less than 1% of their total number of webpages visited during the first two waves of the survey; for 45% of respondents, it makes up between 1-5% of their site visits during this time period, and for 18% of respondents it makes up between 5-10% of their visits.[29]

Figure 2: Distribution of news consumption across respondents



The boxplots show the total number of visits by each respondent to a variety of types of news sites during the first two waves of the survey, along with the proportion of news URLs visited by each respondent relative to the total number of URLs, both of which are displayed on a log scale. The plots show that although some Americans are news junkies, news makes up a relatively small proportion of most respondents' online activity: for half of respondents, news makes up less than 1% of their total page visits, and for a third of respondents, it makes up between 1-5% of their page visits.

We then turn to the question of how much news about trade respondents accessed. Of the unique 142,841 articles respondents accessed from hard news sites for which we have data, roughly 4.8% of them mentioned trade.[30] Figure 3 shows just how far removed news about trade is to most

[29]Note, however, that our measure of news consumption requires respondents to visit one of these 500 domains; respondents who encounter the headlines of news articles while scrolling through their Facebook or Twitter feed, for example, but who do not click on the links to actually access the articles themselves, will not be included here. See Anspach, Jennings and Arceneaux (2019). Our low measure of news consumption is similar to that obtained by other researchers using behavioral measures; see, e.g. Allen et al. (2020); Stier et al. (2022).

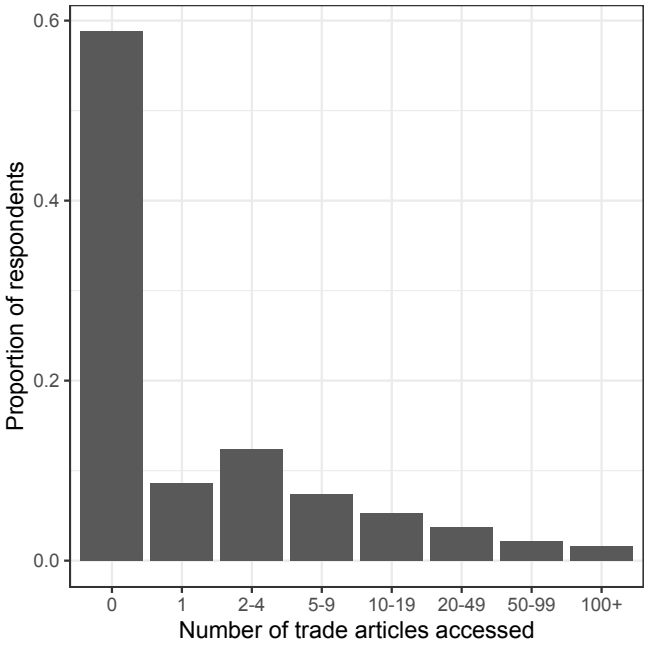
[30]Respondents accessed 152,922 unique hard news URLs in total; we were able to successfully scrape 142,841 of them.

Table 1: Exposure to trade news

	Binary (OLS)			Count (negative binomial)		
	(1)	(2)	(3)	(4)	(5)	(6)
Some college	0.059	0.023	0.020	−0.113	−0.146	−0.025
	(0.044)	(0.056)	(0.057)	(0.237)	(0.323)	(0.326)
College/university	0.027	−0.006	−0.008	0.094	−0.053	0.079
	(0.040)	(0.051)	(0.052)	(0.213)	(0.303)	(0.307)
Postgraduate	0.057	0.010	0.010	−0.148	−0.258	−0.180
	(0.045)	(0.060)	(0.062)	(0.236)	(0.366)	(0.364)
Male	0.094***	0.099***	0.097***	0.196	0.286	0.320*
	(0.028)	(0.034)	(0.035)	(0.146)	(0.187)	(0.185)
Age	−0.096	−0.150*	−0.168**	−0.314***	−0.413***	−0.417***
	(0.065)	(0.080)	(0.082)	(0.078)	(0.104)	(0.102)
Party ID (Republican)	−0.050	−0.062	−0.058	−0.683***	−0.766***	−0.698*
	(0.038)	(0.046)	(0.064)	(0.206)	(0.287)	(0.392)
White	0.025	0.054	0.053	0.335*	0.157	0.086
	(0.036)	(0.045)	(0.045)	(0.195)	(0.260)	(0.262)
Political interest	0.232***	0.207***	0.199***	2.012***	1.803***	1.606***
	(0.054)	(0.066)	(0.070)	(0.297)	(0.379)	(0.411)
Unemployed		0.070	0.070		0.107	0.100
		(0.092)	(0.093)		(0.099)	(0.096)
log(Export Orientation)		0.015	0.015		0.079	0.189
		(0.027)	(0.027)		(0.229)	(0.227)
log(Import Orientation)		−0.025	−0.026		−0.158	−0.259
		(0.027)	(0.027)		(0.237)	(0.234)
Mean Wage		−0.00000	0.00000		0.077	0.095
		(0.00000)	(0.00000)		(0.106)	(0.110)
Offshorability		−0.0001	−0.0001		0.051	0.029
		(0.001)	(0.001)		(0.104)	(0.105)
RTI		−0.030	−0.025		0.088	0.125
		(0.031)	(0.032)		(0.121)	(0.127)
Isolationism			−0.014			−0.114
			(0.067)			(0.117)
MI			−0.066			−0.328**
			(0.077)			(0.135)
CI			−0.032			−0.302**
			(0.081)			(0.146)
Nat. Attach			0.007			−0.184*
			(0.083)			(0.096)
Nat. Chauv			0.006			0.120
			(0.095)			(0.125)
URLs visited	0.00001***	0.00001***	0.00001***	2.018***	1.887***	1.904***
	(0.00000)	(0.00000)	(0.00000)	(0.178)	(0.222)	(0.221)
Intercept	0.116	0.053	0.141	−0.526	−0.163	−0.114
	(0.073)	(0.126)	(0.175)	(0.342)	(0.506)	(0.528)
N	1,214	818	804	1,214	818	804
Adjusted R^2	0.126	0.135	0.127			
AIC				4,843.547	3,240.223	3,209.058

*p < .1; **p < .05; ***p < .01. Explanatory variables standardized in count models.

Figure 3: Distribution of trade news consumption across respondents



59% of respondents accessed no articles relating to trade during the entire seven waves of the survey; 80% of respondents accessed four articles or less during this period.

Americans; 59% of respondents accessed *zero* articles that made even a passing reference to trade during the entire study; eighty percent of respondents accessed four or fewer. This is true even though our baseline measure of news relating to trade is relatively permissive: it counts any article that mentions the word "trade", "tariff", or "tariffs" in the text, including those that use these words outside of the meaning we intend (articles about sports trades or trade secrets, for example).[31]

Columns 1-3 of Table 1 present a series of linear probability models for whether respondents accessed any trade articles at all; columns 4-6 present a series of negative binomial models where the dependent variable is the number of trade articles each respondent accessed. Since the predictors of trade news exposure may be confounded with those of online activity more generally, in both cases we also control for the total number of URLs respondents visited over the course of the study.[32] In the LPMs in models 1-3, men are more likely to be exposed to articles about trade than women (consistent with Burgoon and Hiscox, 2004), and older respondents are less likely to be exposed

[31] To estimate this baseline measure's false positive rate, we hand-code a random sample of 267 articles that mention at least one of the three keywords, finding 28 percent of them do not use the keyword in a manner relevant to international trade. If we assume similar false positive rate holds for our data as a whole, this implies that 70% of respondents accessed no articles relating to trade during our study.

[32] Explanatory variables have been standardized for the negative binomial models to facilitate model convergence.

than younger respondents. The substantively largest effect belongs to self-reported measures of political interest: the politically engaged are 20 - 23 percentage points more likely to be exposed to news about trade than those who aren't. In the negative binomial models in columns 4-6, we find older respondents tend to access fewer online trade news articles than younger respondents, while party ID is also a consistently significant predictor, with Republicans consuming fewer trade articles than Democrats, even controlling for a range of demographic and dispositional covariates. As before, however, the substantively largest effects belong to self-reported political interest. Table B1 in Appendix §B shows that many of the results above hold for exposure to news online as a whole, and are not unique to news about trade (in particular, older respondents consume less news online than younger respondents do, and respondents higher in political interest consume more news than respondents who are less interested in politics). Interestingly, there's some evidence that unemployed individuals consume more news online, although they also display more online activity in general in our data.

Consistent with Guisinger (2017), then, these findings show that most Americans consume relatively little news about trade the way we measure it here. This is the case even though these seven survey waves coincide with NAFTA renegotiations, steel tariffs against Canada, Mexico, and the European Union, and an escalating trade war with China, and even our own respondents suggest that trade is one of the more important problems the country is facing, with nearly a third of respondents during the second wave selecting "international trade imbalances" from a list as something they consider to be particularly important in the United States.[33] This is also the case even though online news is often argued to increase the likelihood of incidental exposure, as individuals unintentionally encounter news surfing the web that they wouldn't intentionally seek out (Tewksbury, Weaver and Maddex, 2001; Barberá, 2014).

5.2 What information about trade do individuals see?

For a finer-grained measure of trade news less susceptible to concerns about false positives, we turn to topic modeling, an unsupervised machine learning technique for the automated content analysis of text (Quinn et al., 2010; Grimmer and Stewart, 2013). We estimate a 100-topic Latent Dirichelet Allocation (LDA) model on the 6894 trade articles respondents accessed during the study.[34] 10 of

[33] As Table B2 in Appendix §B shows, respondents who see trade as an important issue tend to be Republicans, hawks, and nationalists.

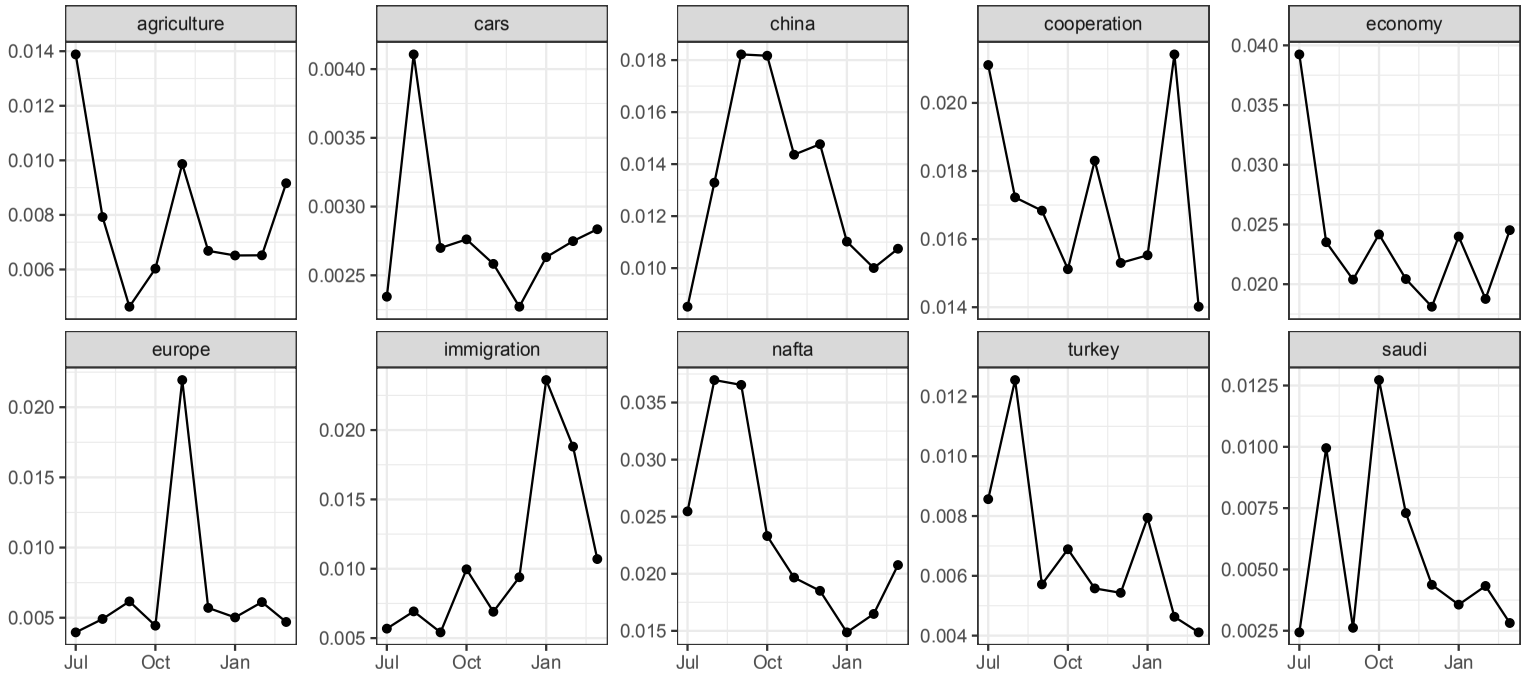
[34] See Appendix §B.2 for model fit statistics.

the topics are semantically interpretable while also featuring discussions of trade. Sample articles associated with each topic are excerpted in Appendix §B.2, and month-level topic probabilities are shown in Figure 4. Since these analyses are specifically on the 6894 articles respondents accessed that mention trade, rather than all 150,000 news articles respondents accessed, the figure can be interpreted as showcasing the wide variety of contexts in which respondents encountered information about trade: from articles about agricultural issues (such as trade disputes over US soybeans, for example), to NAFTA re-negotiations between the United States, Canada and Mexico, to the trade war with China more generally. The spikes in the topic probabilities in Figure 4 closely follow external events, such as the surge in attention to NAFTA with the USMCA negotiations at the end of August, President Trump claiming a new trade deal with Mexico will pay for a border wall in January, or the spike in attention to articles that mention trade in the context of Europe, coinciding with the 100th anniversary of the end of the First World War.

We measure exposure to each of these topics in a variety of ways. The first is in terms of Counts. As Figure 5(a) shows, the topic model classifies the probability that each article is associated with each topic; to produce the count variables, we classify each article based upon its highest probability topic, and calculate the number of URLs visited by each respondent in each topic. The second is in terms of Exposure. Rather than classify each article as belonging only to one topic, for our exposure measures we calculate the sum of probabilities that each article is classified as being in a given topic, for all trade articles visited by that respondent. The third is in terms of Sentiment. We use the Lexicoder sentiment dictionary (Young and Soroka, 2012) to calculate the average sentiment of each of these trade articles, computing sentiment based on an analysis of the 100 words before and after the mention of each of our three trade keywords, to capture whether trade is discussed in a positive or negative fashion. We also calculate sentiment scores at the topic level, classifying each article by the highest probability topic as in the Count measure described above. As Figure 5(b) shows, trade coverage of some topics (like cars) during this period skew more positive, whereas others (such as Turkey) skew more negative, and there is considerable variation in the sentiment expressed in trade coverage overall.[35] Since our baseline trade exposure measure likely contains false positives as noted above, in Appendix §B.2 we also use these topic model results for more restrictive measures of trade news, where we first focus on those articles where the highest probability topic is

[35]Some caution should be taken in the interpretation of these topic-level results given the relatively small number of articles associated with certain topics. As a result, the analysis below largely focuses on the effects of exposure at the aggregate level rather than the topic level.

Figure 4: Evolution in topic attention over time

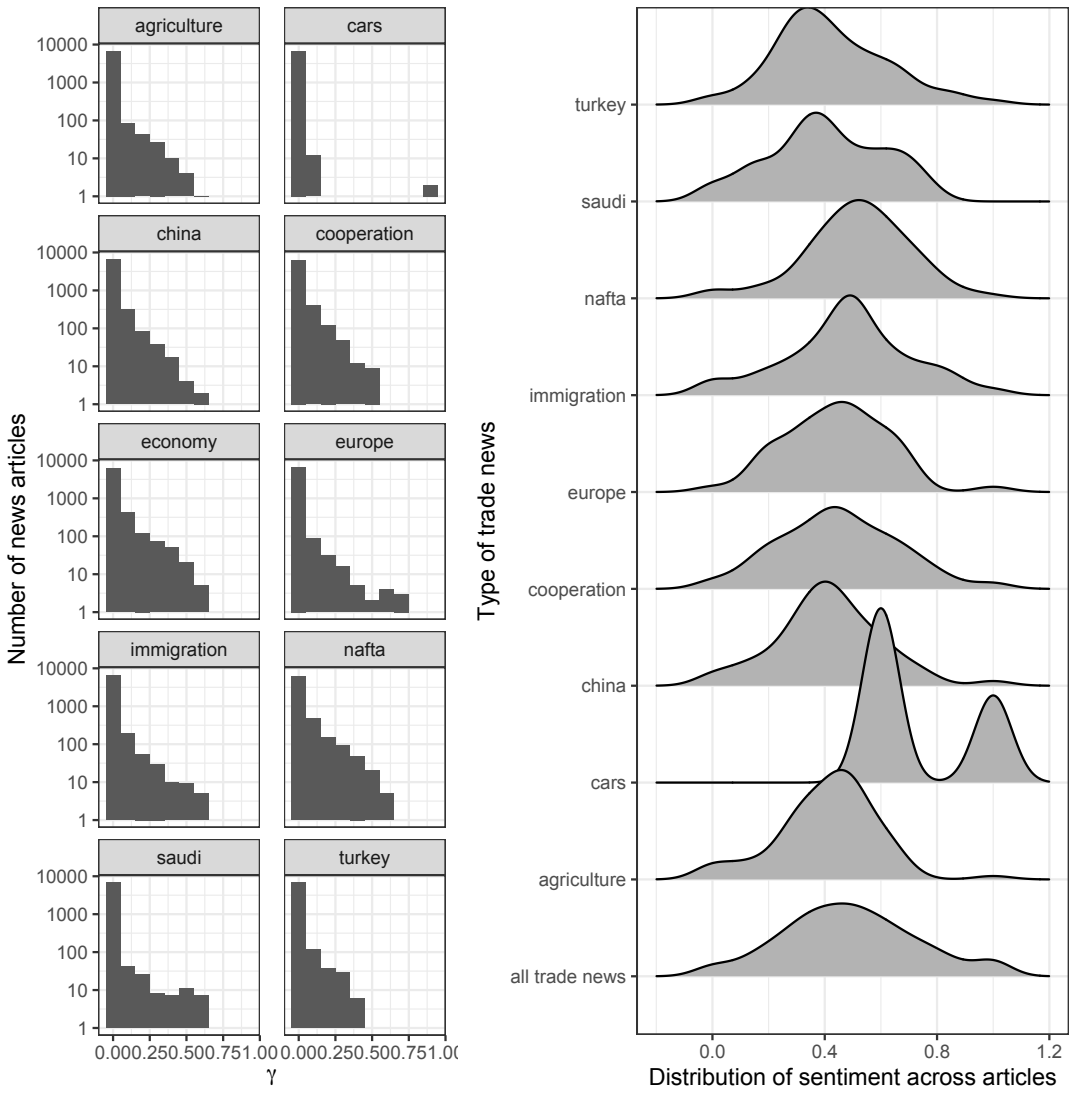


Month-level topic probabilities for each of the 10 topics track with external events (e.g. USMCA negotiations in August 2018 for the NAFTA topic).

Figure 5: Distribution of probabilities and sentiment for each trade topic

(a) Topic probabilities (γ)

(b) Sentiment scores



In the left-hand panel, higher values of γ indicate a higher probability that each article is associated with a given topic. In the right-hand panel, higher sentiment scores indicate more positive sentiment towards trade as measured by the Lexicoder sentiment dictionary, computing sentiment based on an analysis of the 100 words preceding and following the mention of each of our three trade keywords (trade, tariff, tariffs) in each article. See Appendix §B.2 for article excerpts for each topic.

one of the topics presented in Figure 5, and then only retain articles from the subset of topics we classify as the most theoretically relevant to trade.

5.3 Testing the pocketbook mechanism

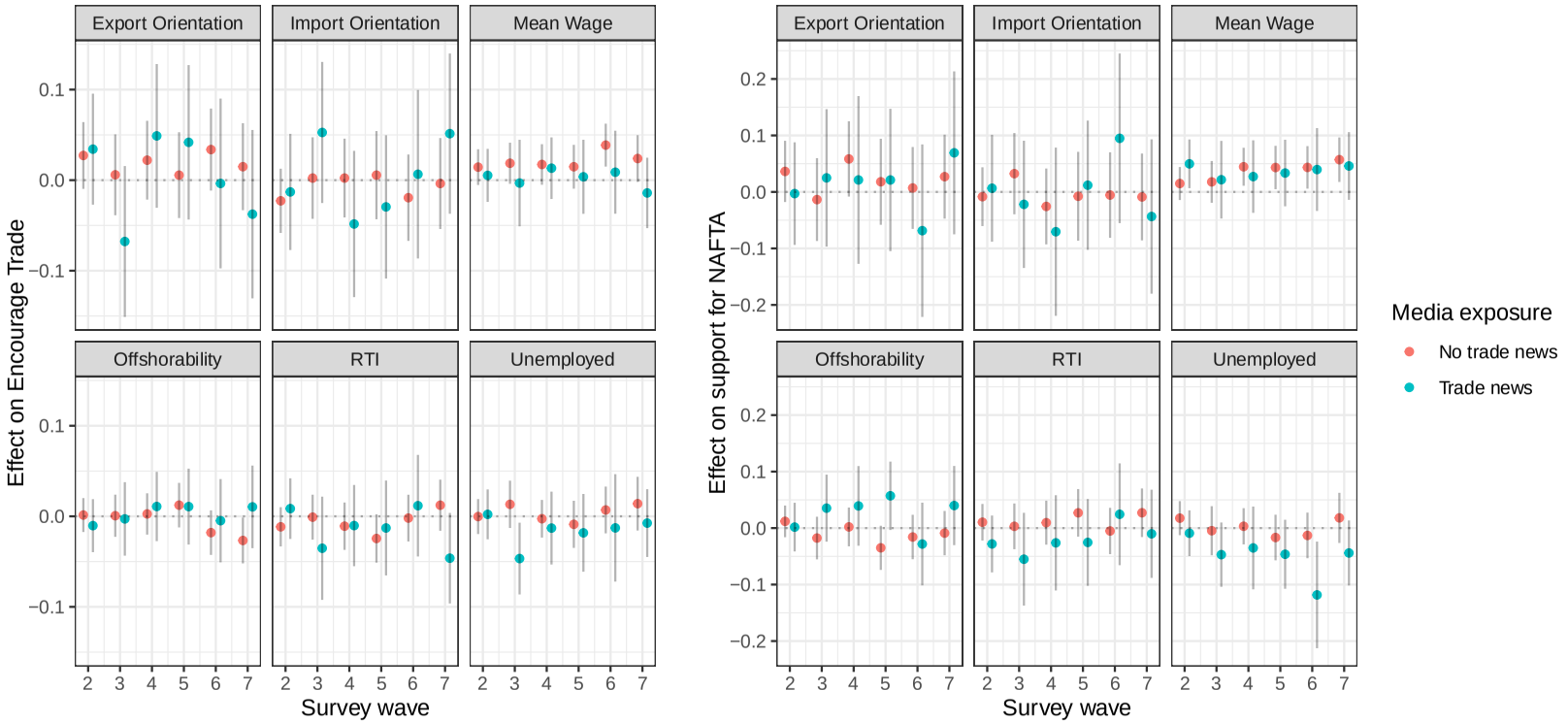
If information about trade affects trade preferences by changing respondents' evaluative beliefs about trade's effects on pocketbook considerations, we should expect that respondents exposed to more news about trade will display trade preferences that better reflect their economic interests than respondents exposed to less news about trade. We test this hypothesis in two ways. First, we estimate a series of linear regression models, regressing respondents' trade attitudes in a given wave on their economic interests and basic demographic covariates, estimating separate models for individuals who were and were not exposed to news about trade during the preceding survey wave.[36] Figure 6 replicates previous work by Mansfield and Mutz (2009) and others in showing that individual-level economic interests generally fail to predict trade preferences, but importantly, it also shows that respondents who were exposed to news about trade did not tend to display stronger relationships between their economic interests and their trade preferences than respondents who were not exposed to news about trade. The one exception appears to be for unemployment, which displays a significant negative effect on encouraging trade (in wave 3) and on support for NAFTA (in wave 6), but the estimates are noisy due to the relatively low number of unemployed individuals in the data.[37] The analysis here defines exposure to trade news as accessing a single article about trade during the preceding survey wave, which would place respondents in the 81st-86th percentile of trade news consumption in our sample, depending on the wave, but Appendix §B shows that similar results obtain when we use a higher threshold for what counts as exposure to trade news. Likewise, Appendix §B.2 shows that we continue to obtain null results when we use the topic model to restrict our measure of trade news to those articles where the highest probability topic is one of the 10 topics presented in Figure 4, as well as further restricting the analysis by only looking at the topics of the most theoretical importance for the study of trade attitudes.

In Figure 7, we illustrate these results another way, presenting the marginal effects of each of our

[36] Given the number of different ways economic interests can manifest themselves, subsetting the sample in this manner allows us to avoid a high-dimensional interactive model. The industry and occupation open-ended response items used to generate measures of economic interests were administered in the first two survey waves, but largely remained constant. In the analysis below, we use the measure of economic interests from the first wave as our covariates of interest, although the results remain the same when we use the measure from the second wave.

[37] The highest level of unemployment reported in our data is in the first wave of the study, when 4% of respondents are unemployed.

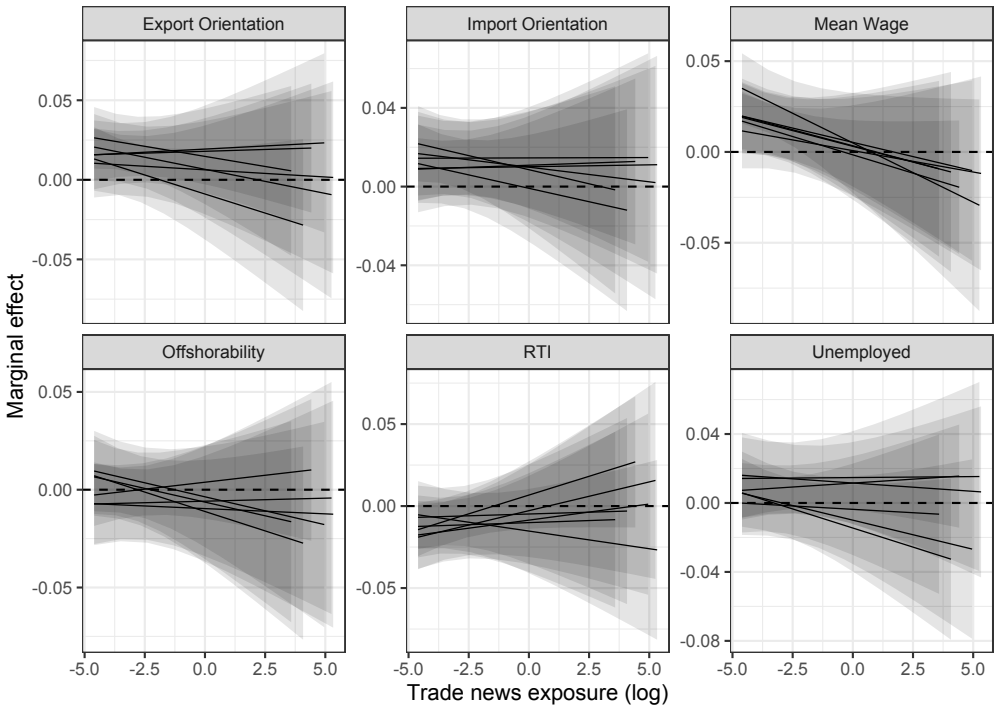
Figure 6: Are economic interests more likely to shape trade preferences when individuals are exposed to news about trade?



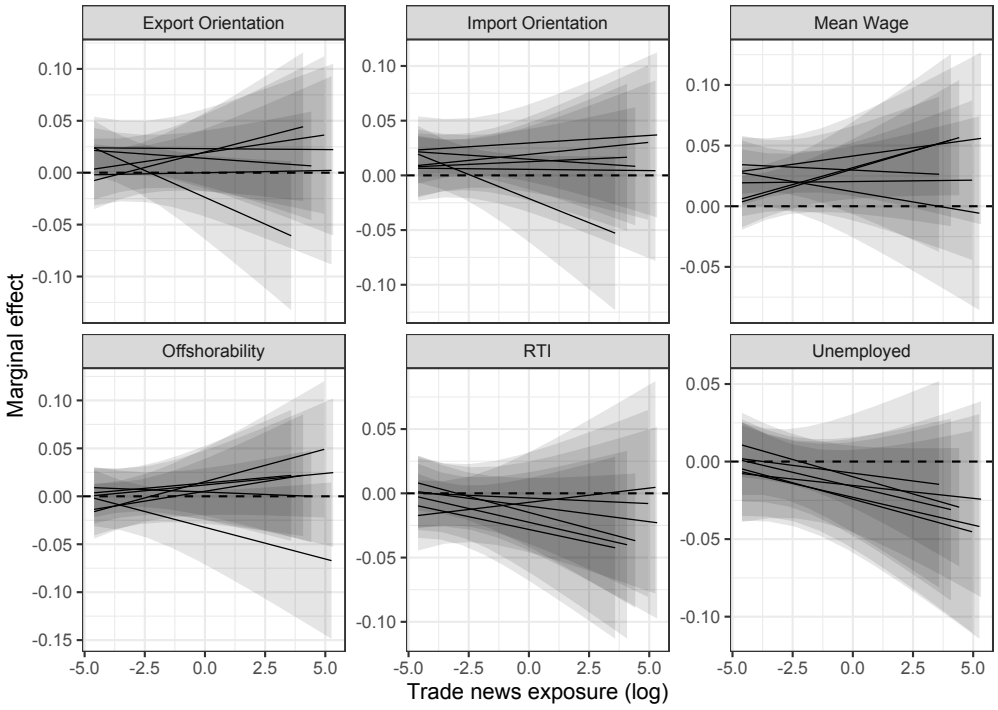
Respondents exposed to trade news (in turquoise) are no more likely to express trade preferences consistent with their economic interests than respondents who were not exposed to trade news (in red). The analysis here defines exposure to trade news as accessing a single article about trade during the preceding survey wave, which would place respondents in the 81st-86th percentile of trade news consumption in our sample, depending on the wave. See Appendix §B for a supplementary analysis showing the results hold with a higher threshold for exposure to trade news.

Figure 7: Marginal effects of economic interests do not increase with trade news exposure

(a) Encourage Trade



(b) NAFTA



See Appendix §B for a supplementary analysis using finite mixture models.

pocketbook considerations conditioned on a continuous measure of trade news exposure, controlling for a range of demographic and dispositional characteristics. We once again find little evidence that exposure to trade news amplifies the effects of economic interests on trade preferences for either of our trade attitude measures, in any of our study waves. The one exception is for the effects of mean industry wage, where it appears that exposure to more news about trade may weaken the effect of industry wage on support for encouraging trade – although here the findings are the opposite of what pocketbook-based models of economic interests would predict, which would lead us to expect that respondents in high-wage industries in the United States should become more supportive about trade the more news about trade they encounter.

Moreover, in Appendix §B we further replicate these results using a set of finite mixture models modeling trade preferences as a function of either economic interest or ideational variables, modeling the mixing probability as a function of exposure to trade news. We once again find little evidence in favor of the claim that respondents who consume more news about trade are more likely to anchor their trade preferences on economic interests, and show that between 64-82% of the observations (depending on the wave) are better explained by ideational variables than by economic interest variables. In Appendix §B.4 we replicate the above analyses, but including community-level measures of economic interests in addition to individual-level measures (Frieden, 2022). We find once again that trade attitudes are better explained by ideational variables than by economic interest measures, and that exposure to news about trade does not moderate the effects of these community-level characteristics.

5.4 Testing the sociotropic mechanism

Although there appears to be little evidence that the effect of news about trade shapes trade preferences through pocketbook pathways, there is another potential mechanism, emphasizing sociotropic considerations instead. If information about trade affects trade preferences by changing respondents' evaluative beliefs about trade's effects on their broader community, trade preferences will be associated with the valence of information exposure more generally: the more positive stories about trade to which respondents are exposed, the more supportive of trade they'll be, while the more negative stories about trade to which respondents are exposed, the more negative towards trade they'll be. In this sociotropic pathway, news about trade shapes trade preferences by conveying information about whether trade is good or bad more generally, rather than whether trade is good or bad for

individuals specifically like themselves.

To test this mechanism, because our interest is in the effects of trade news exposure over time, rather than the role of variables like economic interests that are effectively static in our data, we can exploit the panel structure of our data to estimate a set of fixed effect regression models of the form $y_{i,t} = \alpha_i + \beta X_{i,t-1} + \epsilon_{i,t}$, where trade attitudes are modeled as a function of trade news exposure in the preceding period, controlling for time-invariant characteristics of respondents. We also estimate an interactive version of the model, $y_{i,t} = \alpha_i + \beta_1 X_{i,t-1} + \beta_2 Z_{i,t-1} + \beta_3 X Z_{i,t-1} + \epsilon_{i,t}$, in which the effect of trade news exposure is conditional on the sentiment about trade expressed in the trade news. This within-subject approach also gives us more leverage in terms of causal identification than the previous set of analyses.[38] Table 2 presents the results for each of our two trade dependent variables. Models 1-3 show that there is no significant main effect of exposure to trade news on subsequent trade preferences, but model 2 finds a significant positive interaction effect between trade news exposure and the average sentiment about trade expressed in the articles.[39] The conditional effect, depicted in Figure 8, shows that for those respondents who did access trade articles in the previous time period, support for free trade increases when respondents are exposed to more positive stories about trade. Supplementary analyses in Appendix §B, using more restrictive measures of trade news derived from topic models, similarly show that respondents exposed to positive news about trade are significantly more supportive of trade than respondents exposed to negative news about trade; the analysis in Table B4 in Appendix §B suggest that the effect is driven by exposure to positive news about trade bolstering support for trade, rather than exposure to negative news about trade dampening support. And, in Appendix §B.5, we replicate the above findings using a supervised machine learning classifier trained on news articles that feature trade-related keyword tags published by the news outlets themselves, showing the above findings hold.

However, there are two caveats worth mentioning. First, using all of our measures of trade news exposure, we only find a significant effect on support for free trade in general, rather than NAFTA attitudes in particular. Second, as the low R^2 statistics suggest, it appears that trade news

[38] The panel regressions are not susceptible to bias due to omitted time-invariant factors, although they too rely on assumptions in order to support a causal interpretation. The most important threat to inference in this setting is the possibility of trade news exposure affecting time-varying confounders (for example, news interest). Such pathways could bias estimates if not adequately specified, although the time span of our data presumably rules out large indirect effects of this kind.

[39] Since the sentiment score captures the average sentiment about trade a respondent was exposed to in all of the articles they accessed, models 2 and 4 restrict the analysis only to those respondents who accessed trade articles. See Appendix §B for analysis comparing respondents exposed to positive trade news, negative trade news, and no trade news.

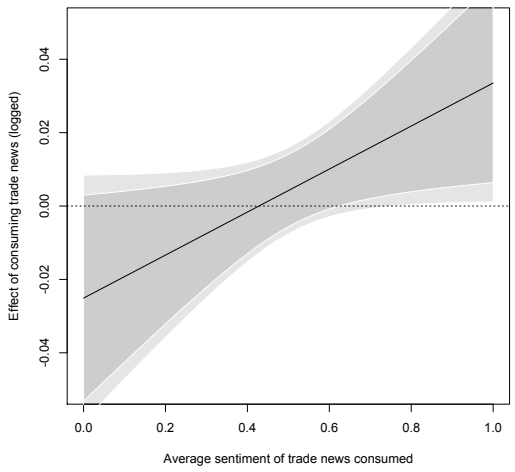
exposure generally does a poor job of explaining within-subject change in trade attitudes among our respondents, even with our fairly fine-grained measures of trade news exposure.

Table 2: Does exposure to news about trade affect trade preferences?

	Encourage Trade		NAFTA	
	(1)	(2)	(3)	(4)
Trade news exposure (logged)	0.00001	−0.025	−0.002	−0.004
	(0.001)	(0.017)	(0.002)	(0.022)
Trade news sentiment		0.010		0.036
		(0.036)		(0.045)
Trade news exposure x sentiment		0.059*		0.010
		(0.031)		(0.040)
Respondent FEs	✓	✓	✓	✓
N	4,819	1,206	4,617	1,168
R^2	0.000	0.010	0.001	0.002

*p < .1; **p < .05; ***p < .01

Figure 8: Consuming more positive news stories about trade increases support for encouraging trade



6 Conclusion

IPE scholars seeking to explain why economic interests fail to predict trade preferences have increasingly turned to the role of information, fielding innovative experimental designs that randomly present respondents with information about trade, and measure its effects on respondents' subsequent trade preferences (e.g. Rho and Tomz, 2017; Schaffer and Spilker, 2019; Jamal and Milner, 2019). In this paper, we seek to build on these important studies, while asking a slightly different question: what kind of information about trade are citizens exposed to in the real world? How does information shape trade preferences in the wild?

We investigate these questions by fielding a seven-wave panel survey on a nationally representative sample of Americans over a 13 month period, where we measure respondents' economic interests, track their trade preferences, and crucially, obtain unobtrusive behavioral measures of their media consumption during this time period. By reconstructing respondents' online information environments at the URL level, we are able to measure how much news about trade citizens are exposed to in a naturalistic setting, and test a variety of competing theories about how information shapes trade preferences. Consistent with theories of economic ignorance, we show how little news about trade most Americans consume. However, because the kind of information about trade respondents encounter in the real world differs from the types of information respondents are presented with in stylized survey experiments, we find little evidence that information about trade shapes trade preferences through pocketbook-based pathways, magnifying the effects of economic interests. Instead, we find some evidence in favor of information about trade shaping trade preferences through sociotropic pathways: the more positive news about trade respondents access, the more supportive of trade they become. While our findings were obtained during a different trade war, in a somewhat different media environment, we have little theoretical reason to believe that the sociotropic mechanisms we discuss here are temporally bounded, and that contemporary media coverage is more amenable to pocketbook-based explanations. Nonetheless, future scholars should utilize digital trace data to replicate our findings in other contexts.

These findings have a number of important implications. For IPE scholars, they point to the continued importance of sociotropic rather than pocketbook-based pathways when understanding the origins of economic preferences, echoing previous work about the limitations of self-interest (Mansbridge, 1990; Mansfield and Mutz, 2009; Hainmueller and Hiscox, 2010). For public opinion and

communications scholars, they show the value of obtaining behavioral rather than self-reported measures of media consumption. Echoing other work demonstrating the low salience of trade (Guisinger, 2009), they also show how little information most Americans receive about trade: even during a year of high stakes trade wars and highly salient NAFTA renegotiations, a majority of our respondents appear to not have been exposed online to news mentioning trade. This raises questions for voter-driven models of IPE, as well as scholars seeking to explain why democracies conduct systematically different trade policies than non-democracies (Mansfield, Milner and Rosendorff, 2000; Kono, 2006).

For political scientists more broadly, they suggest that understanding the origins of preferences may often require going outside of the lab and into the wild. Experimental methods are valuable because of their ability to identify causal effects, but they identify the effects of causes, rather than the causes of effects — which makes their growing popularity in IPE for studying the origins of preferences not without its challenges. This is somewhat different from usual external validity-based critiques about experimental methods in political science (McDermott, 2011): the concern is not that treatments would display a different effect on other populations or outside the confines of the lab, but rather, that respondents in the real world may not receive the treatment (Althaus et al., 2011), or receive a different type of treatment altogether. Our study therefore has implications for scholars fielding experiments more broadly, speaking to larger questions about how descriptive work can productively guide experimental designs.

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Trade Attitudes In The Wild
Supplementary Appendix

October 7, 2025

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A Sample information

Respondents were sampled according to YouGov's demographic/political targets, and then reweighted in order to obtain a sample that is representative of the U.S. population using weights computed by YouGov. As stated by YouGov, respondents were "were weighted according to a sampling frame constructed by stratified sampling from the full 2016 American Community Survey (ACS) 1-year sample with selection within strata by weighted sampling with replacements (using the person weights on the public use file). The sample cases were weighted to the sampling frame using propensity scores. The sample cases and the frame were combined and a logistic regression was estimated for inclusion in the frame. The propensity score function included age, gender, race/ethnicity, years of education, and region. The propensity scores were grouped into deciles of the estimated propensity score in the frame and post-stratified according to these deciles. The weights were then post-stratified on 2016 Presidential vote choice, and a four-way stratification of gender, age (4-categories), race (4- categories), and education (4-categories), to produce the final weight."

A.1 Additional information about Pulse

Although the type of web tracking data using Pulse that we employ here is new in the study of IR, it has been fruitfully utilized in a variety of other political science applications (e.g. Guess, 2021; Stier et al., 2022; Munzert et al., 2024; see Clemm von Hohenberg et al., 2024 for a discussion of best practices). Figure A1 and Tables A1-A2 compare responses on questions about online privacy between the general YouGov population (both with and without Pulse: $N = 1000$) and for respondents in the general Wakoopa panel ($N = 6591$), showing that individuals who consent to passive webtracking are less concerned about their data online than the population as a whole. Table A1 codes respondents with no Pulse activity for more than a year as not a Pulse member, whereas Table A2 codes all respondents who joined at any time as a Pulse member, regardless of the recency of their Pulse activity. Interestingly, in a multivariate context Pulse members only differ from non-members in terms of their concerns about data, rather than their beliefs about privacy, use of ad blocker software, or concerns about personalized advertising. Guess, Nyhan and Reifler (2018, S2) compare demographic distributions between the 2016 American National Election Studies (ANES) pre-election face-to-face study, and YouGov Pulse panel members from that year, showing the distributions are largely similar.

There are a variety of other limitations in the data worth flagging. As noted in the discussion of ethical considerations below, respondents are participating in the study with fully informed consent, and have the

option of turning the software on and off at any time. As with any web tracking data, the granularity of the data provided also depends on respondents' browser and operating system. Bosch et al. (2025) find that digital trace data is often susceptible to undercoverage (primarily due to undercoverage across devices), which might imply our estimates of exposure to news have a downward bias. The data collected are thus nonetheless inevitably incomplete – although we do not have reasons to believe respondents are particularly likely to turn off the tracking software in order to clandestinely read news articles about trade, and an examination of the sites accessed by many respondents suggest relatively little concern about social desirability.

Table A1: Pulse Membership and Internet Privacy Attitudes (1)

	Concerned About Data	Privacy Doesn't Matter	Ad Blocker	Ads Scare Me
	(1)	(2)	(3)	(4)
Pulse	−0.239*	0.094	−0.001	−0.205
	(0.139)	(0.131)	(0.171)	(0.142)
Age	0.002	−0.007**	−0.005	−0.002
	(0.003)	(0.003)	(0.003)	(0.003)
White	−0.113	−0.286*	−0.168	0.124
	(0.150)	(0.156)	(0.162)	(0.175)
Black	−0.391*	−0.054	−0.454**	0.035
	(0.232)	(0.215)	(0.229)	(0.236)
Hispanic	−0.091	0.117	−0.307	0.112
	(0.220)	(0.244)	(0.252)	(0.243)
Male	0.064	0.001	0.214**	−0.169*
	(0.093)	(0.091)	(0.108)	(0.092)
Income: < $50k	−0.293**	0.218	−0.270*	−0.130
	(0.120)	(0.136)	(0.154)	(0.127)
Income: $50-100k	−0.315**	0.260*	−0.245	−0.106
	(0.123)	(0.139)	(0.150)	(0.133)
Income: $100k+	−0.368***	0.101	−0.304*	−0.260
	(0.140)	(0.141)	(0.180)	(0.167)
Some college	0.098	−0.210*	0.092	0.050
	(0.119)	(0.113)	(0.132)	(0.116)
College graduate	0.179	−0.121	0.259*	0.104
	(0.140)	(0.117)	(0.137)	(0.138)
Postgrad	0.253*	−0.236	0.330*	−0.096
	(0.130)	(0.147)	(0.188)	(0.181)
Constant	3.882***	2.383***	3.891***	3.423***
	(0.222)	(0.206)	(0.255)	(0.225)
N	986	987	988	982
Adjusted R^2	0.027	0.043	0.027	0.004

*p < .1; **p < .05; ***p < .01
OLS models with HC2 robust standard errors and YouGov's sample-matching weights applied.

Figure A1: Pulse respondents differ in their attitudes towards internet privacy

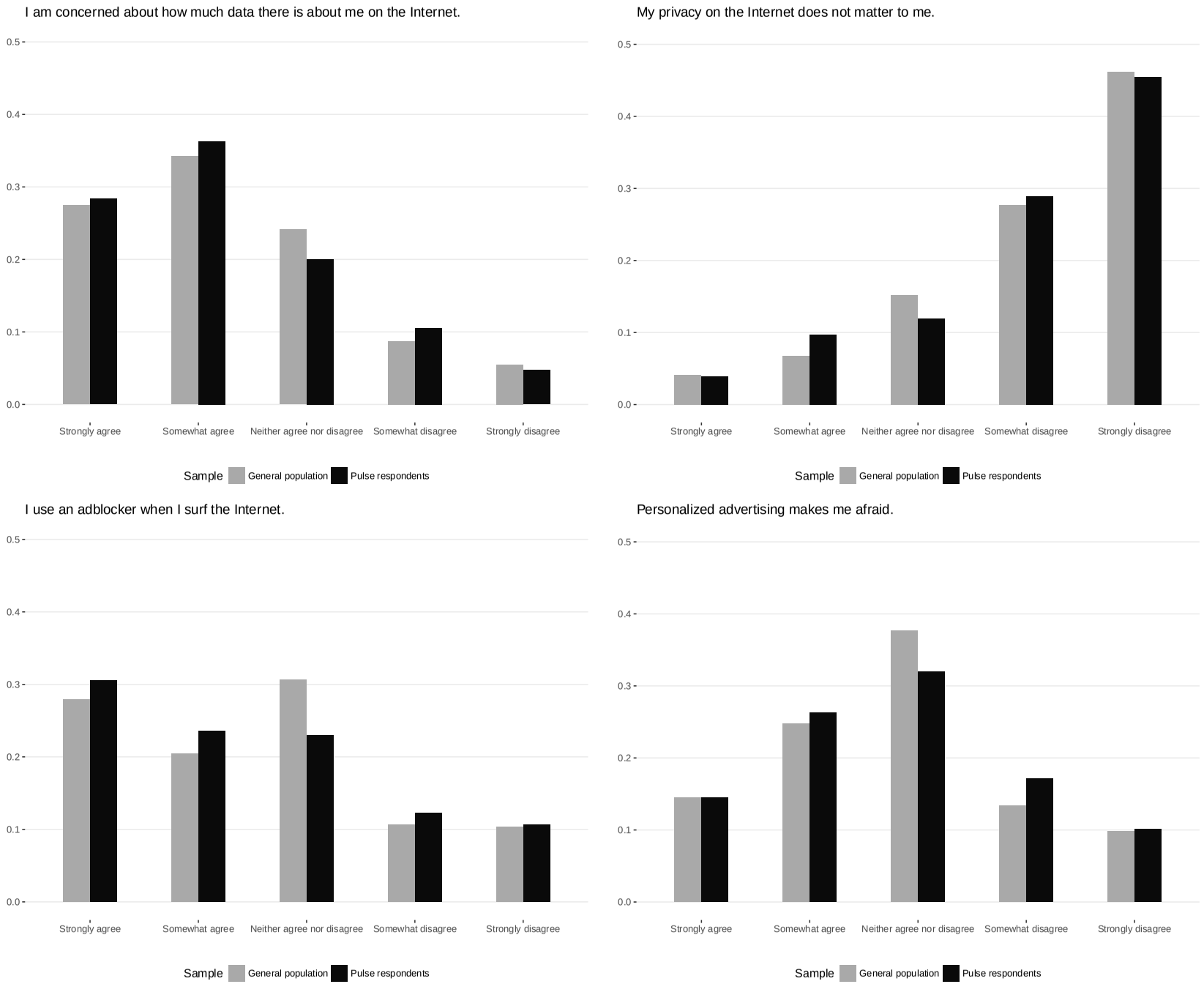


Table A2: Pulse Membership and Internet Privacy Attitudes (2)

	Concerned About Data	Privacy Doesn't Matter	Ad Blocker	Ads Scare Me
	(1)	(2)	(3)	(4)
Pulse	−0.269**	0.133	0.117	−0.089
	(0.127)	(0.122)	(0.144)	(0.127)
Age	0.002	−0.007**	−0.004	−0.002
	(0.003)	(0.003)	(0.003)	(0.003)
White	−0.125	−0.281*	−0.170	0.110
	(0.152)	(0.157)	(0.161)	(0.175)
Black	−0.393*	−0.056	−0.464**	0.021
	(0.232)	(0.216)	(0.231)	(0.235)
Hispanic	−0.107	0.124	−0.303	0.103
	(0.222)	(0.245)	(0.252)	(0.244)
Male	0.068	−0.002	0.207*	−0.175*
	(0.094)	(0.092)	(0.108)	(0.093)
Income: < $50k	−0.281**	0.213	−0.274*	−0.125
	(0.121)	(0.137)	(0.155)	(0.129)
Income: $50-100k	−0.316**	0.261*	−0.244	−0.106
	(0.123)	(0.140)	(0.150)	(0.133)
Income: $100k+	−0.361***	0.098	−0.308*	−0.259
	(0.140)	(0.141)	(0.180)	(0.168)
Some college	0.095	−0.209*	0.092	0.047
	(0.119)	(0.114)	(0.132)	(0.116)
College graduate	0.172	−0.116	0.269*	0.110
	(0.140)	(0.117)	(0.138)	(0.139)
Postgrad	0.249*	−0.235	0.328*	−0.102
	(0.130)	(0.146)	(0.187)	(0.181)
Constant	3.908***	2.364***	3.856***	3.402***
	(0.225)	(0.207)	(0.256)	(0.228)
N	986	987	988	982
Adjusted R^2	0.030	0.044	0.029	0.002

*p < .1; **p < .05; ***p < .01

OLS models with HC2 robust standard errors and YouGov's sample-matching weights applied.

A.2 Consent and ethical considerations

Collecting passive behavioral data that can be linked to survey responses poses privacy and ethical challenges beyond those typically associated with traditional social science research methods (Stier et al., 2019). We join other researchers in striving to meet these challenges with robust procedures designed to ensure that informed consent is obtained, privacy is protected, and personally identifiable information is not stored or reported. For the data analyzed in this study, we followed a strict protocol informed by IRB guidance and emerging best practices (Menchen-Trevino, 2018).[1]

First, as with any survey research, respondents are anonymous to the researchers. While we do not ask for identifying information, it is possible that some subjects visited websites or made search queries that contain personal details (for example, by visiting one's own Facebook profile page). While we did not collect the content of these pages, the URLs themselves could contain such information.[2] We handle this by keeping the raw web visit data stored securely in a strictly separate location from the survey data and by performing the primary analyses for the paper using only individual-level variables that are aggregated to the respondent level (as with counts or shares of visits). Only these aggregates will be released as replication data.

Second, sharing web visits with researchers is done with fully informed consent. People who join the Pulse panel are told about the nature of the data collected, that it is kept anonymous, and that data is not shared with third parties. They are also told that the software can be removed at any time or paused temporarily. When subjects join the Pulse panel, they are given a clear privacy policy that says, in part, that they agree to collection of "the following details on your internet behaviour: the URLs of the websites you visit, what time you visited the websites and the length of time you spent on them, the search words you entered, the software and applications you used and the advertisements you viewed, a unique user number" in addition to "supplementary information from the Panel provider, including the socio-demographic details that you furnished to the Panel provider [YouGov] for this purpose, like your education and age. These details will never contain your name or email address." Before joining this specific study, respondents additionally agreed to a separate consent statement informing them, "Your participation is voluntary. Participation involves completion of a short survey and voluntary tracking of online media consumption. You may choose not to answer any or all questions. Furthermore, you are free to opt out of web tracking, which you may have previously agreed to participate in as part of the YouGov Pulse panel, at any time."

[1] The study was approved by the Institutional Review Board (IRB) at [redacted].

[2] Secure transactions and passwords are not collected or shared with researchers in the first place, and YouGov performs an initial scrub of personally identifying information before delivering the data.

Finally, in contrast to internet users' daily experience on the internet, Pulse panelists are compensated for sharing their data. As researchers become more aware of the risks and benefits of working with linked survey and digital trace data, we hope that these emerging best practices can help to inform a consensus.[3]

B Supplementary analysis

[3]See https://www.aaas.org/news/apsa-releases-statement-research-integrity for the APSA Statement on Field Experiments and Research Integrity in Political Science for an example of how this has worked for researchers who employ data and methods with distinct but serious ethical considerations.

Table B1: Exposure to...

	All news sites				All websites			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Some college	−0.064	−0.460**	−0.455*	−0.449*	0.088	−0.266	−0.242	−0.298
	(0.182)	(0.234)	(0.235)	(0.234)	(0.144)	(0.189)	(0.190)	(0.189)
College/university	−0.006	−0.317	−0.362*	−0.379*	0.281**	0.177	0.118	0.124
	(0.165)	(0.212)	(0.213)	(0.213)	(0.130)	(0.171)	(0.172)	(0.172)
Postgraduate	0.056	−0.301	−0.359	−0.333	0.053	−0.027	−0.056	−0.078
	(0.188)	(0.251)	(0.254)	(0.253)	(0.149)	(0.202)	(0.205)	(0.205)
Male	0.056	0.129	0.163	0.110	−0.130	0.023	−0.047	−0.017
	(0.115)	(0.143)	(0.145)	(0.145)	(0.091)	(0.115)	(0.117)	(0.117)
Birth year	−0.210***	−0.301***	−0.164*	−0.336***	0.148***	0.113*	0.020	0.103*
	(0.059)	(0.073)	(0.087)	(0.073)	(0.047)	(0.059)	(0.071)	(0.059)
Party ID	−0.178	0.104	0.063	0.324	−0.225*	−0.340**	−0.378**	−0.075
	(0.157)	(0.194)	(0.195)	(0.263)	(0.124)	(0.156)	(0.158)	(0.213)
White	−0.010	0.206	0.229	0.119	0.131	−0.064	−0.062	−0.042
	(0.149)	(0.186)	(0.187)	(0.187)	(0.118)	(0.151)	(0.152)	(0.151)
Political interest	1.558***	1.586***	1.626***	1.383***	0.176	−0.059	−0.022	0.046
	(0.224)	(0.274)	(0.275)	(0.290)	(0.177)	(0.221)	(0.222)	(0.234)
Unemployed		0.124*		0.151**		0.151**		0.133**
		(0.074)		(0.074)		(0.060)		(0.060)
Labor force			−0.122				0.181***	
			(0.087)				(0.070)	
log(Export Orientation)		0.064	0.009	0.070		0.115	0.067	0.085
		(0.151)	(0.153)	(0.150)		(0.122)	(0.124)	(0.122)
log(Import Orientation)		0.043	0.111	0.032		−0.062	−0.016	−0.033
		(0.149)	(0.153)	(0.148)		(0.120)	(0.124)	(0.120)
Mean Wage		−0.089	−0.080	−0.057		0.051	0.040	0.044
		(0.079)	(0.080)	(0.080)		(0.064)	(0.065)	(0.065)
Offshorability		0.136*	0.145*	0.112		−0.015	0.011	−0.014
		(0.076)	(0.077)	(0.076)		(0.061)	(0.062)	(0.062)
RTI		−0.111	−0.088	−0.062		0.122*	0.137*	0.113
		(0.088)	(0.089)	(0.089)		(0.071)	(0.072)	(0.072)
Isolationism				−0.162*				0.024
				(0.083)				(0.067)
MI				−0.161*				0.003
				(0.095)				(0.077)
CI				−0.110				0.174**
				(0.104)				(0.084)
Nat. Attach				−0.077				−0.083
				(0.077)				(0.062)
Nat. Chauv				0.044				0.052
				(0.096)				(0.078)
Intercept	4.147***	4.022***	4.035***	4.202***	8.936***	9.366***	9.375***	9.196***
	(0.247)	(0.332)	(0.334)	(0.360)	(0.195)	(0.268)	(0.270)	(0.291)
N	1,214	818	802	804	1,214	818	802	804
AIC	13,524.850	9,097.518	8,942.520	8,984.039	23,892.310	16,002.010	15,701.520	15,750.120

*p < .1; **p < .05; ***p < .01. Negative binomial models.

Table B2: Trade imbalance most important issue

	(1)	(2)	(3)	(4)	(5)	(6)
Some college	0.073*	0.075	0.071	0.087	0.074	0.070
	(0.041)	(0.051)	(0.052)	(0.055)	(0.055)	(0.055)
College/university	0.024	0.024	0.018	0.040	0.049	0.047
	(0.037)	(0.047)	(0.047)	(0.050)	(0.050)	(0.050)
Postgraduate	−0.042	−0.025	−0.031	−0.006	0.018	0.021
	(0.041)	(0.053)	(0.054)	(0.058)	(0.058)	(0.058)
Male	0.046*	0.033	0.031	0.040	0.027	0.035
	(0.025)	(0.030)	(0.031)	(0.033)	(0.033)	(0.033)
Birth year	−0.056***	−0.056***	−0.066***	−0.047***	−0.041**	−0.037**
	(0.013)	(0.015)	(0.018)	(0.016)	(0.016)	(0.017)
Party ID	0.245***	0.236***	0.231***	0.241***	0.055	0.054
	(0.035)	(0.042)	(0.042)	(0.045)	(0.060)	(0.060)
White	0.009	0.021	0.022	0.019	0.017	0.015
	(0.033)	(0.039)	(0.040)	(0.043)	(0.043)	(0.043)
Unemployed		0.004		0.001	0.001	0.001
		(0.017)		(0.018)	(0.018)	(0.018)
In labor force			0.015			
			(0.019)			
log(Export Orientation)		−0.045	−0.053	−0.050	−0.044	−0.041
		(0.032)	(0.033)	(0.035)	(0.035)	(0.035)
log(Import Orientation)		0.046	0.056*	0.053	0.049	0.048
		(0.032)	(0.033)	(0.035)	(0.035)	(0.035)
Mean Wage		−0.005	−0.009	−0.006	−0.006	−0.007
		(0.015)	(0.015)	(0.019)	(0.019)	(0.019)
Offshorability				−0.028	−0.022	−0.021
				(0.018)	(0.018)	(0.018)
RTI				0.001	−0.002	−0.004
				(0.020)	(0.020)	(0.020)
Isolationism					0.039**	0.038**
					(0.019)	(0.019)
MI					0.061***	0.055**
					(0.020)	(0.022)
CI					−0.035	−0.036
					(0.023)	(0.024)
Nat. Attach						0.042**
						(0.017)
Nat. Chauv						−0.013
						(0.023)
Intercept	0.173***	0.179***	0.182***	0.168***	0.254***	0.253***
	(0.044)	(0.058)	(0.058)	(0.062)	(0.065)	(0.065)
N	1,333	992	974	885	869	869
Adjusted R^2	0.066	0.060	0.063	0.057	0.080	0.084

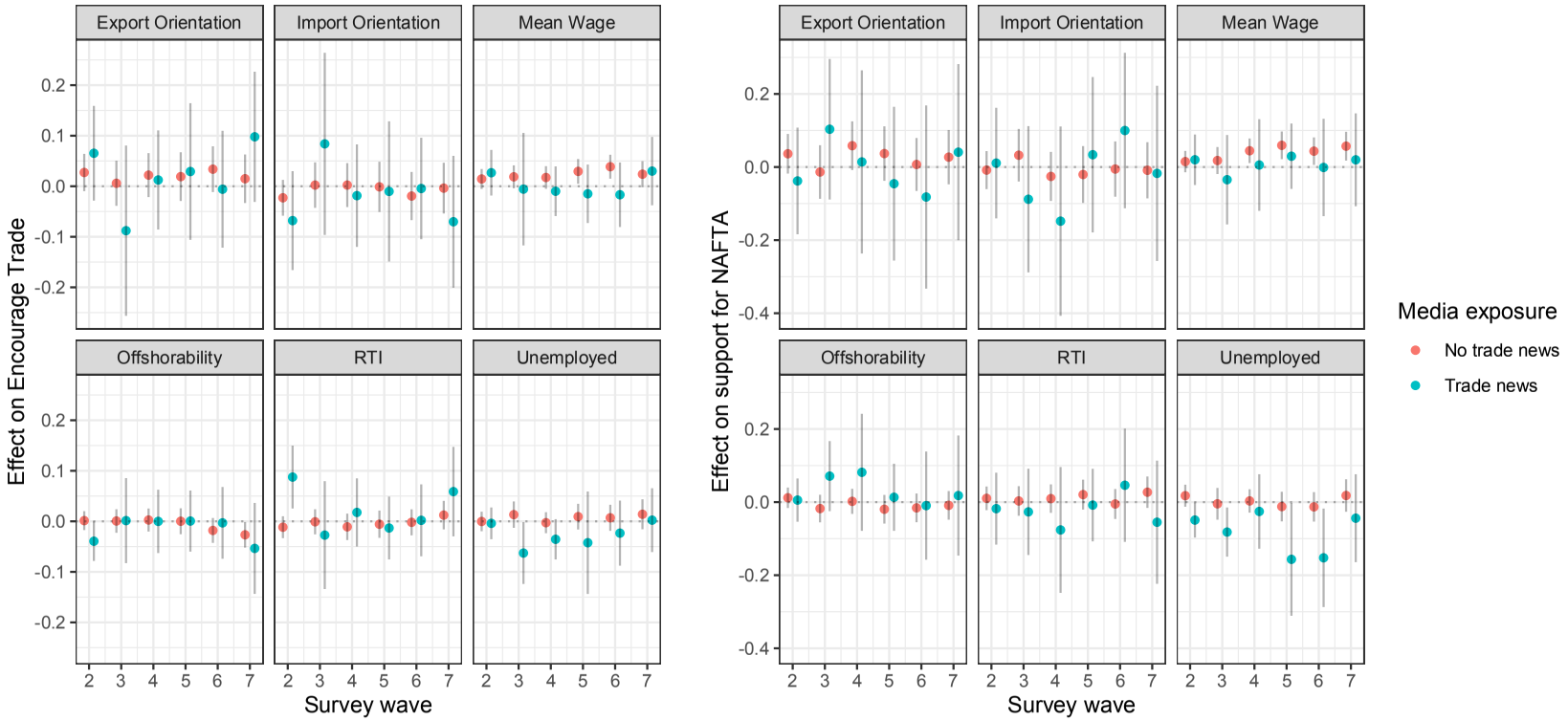
*p < .1; **p < .05; ***p < .01

B.1 Additional pocketbook results

In the main text, we test the pocketbook pathway by estimating a set of marginal effects models, showing that individuals who consume more news about trade do not display stronger relationships between their economic interests and their trade preferences. Here we build on these findings with a set of finite mixture models, which are useful for testing the effects of competing hypotheses, testing the scope conditions in which one set of explanatory variables performs better than another (Imai and Tingley, 2012). Rather than assume that trade preferences are shaped either by material economic interests or by ideational variables, we can model trade preferences as a mixture of the two, estimating the probability that each respondent's trade preferences are consistent with the predictions of one set of variables rather than the other, and using observed covariates about each respondent to help model the conditions under which each set of explanatory variables should better predict trade attitudes. In our case, we model trade preferences as a function either of the economic interest variables presented in the main text, or the ideational variables (militant internationalism, cooperative internationalism, isolationism, national attachment and chauvinism, and party identification) discussed in main text, modeling the mixing probability as a function of exposure to trade news.

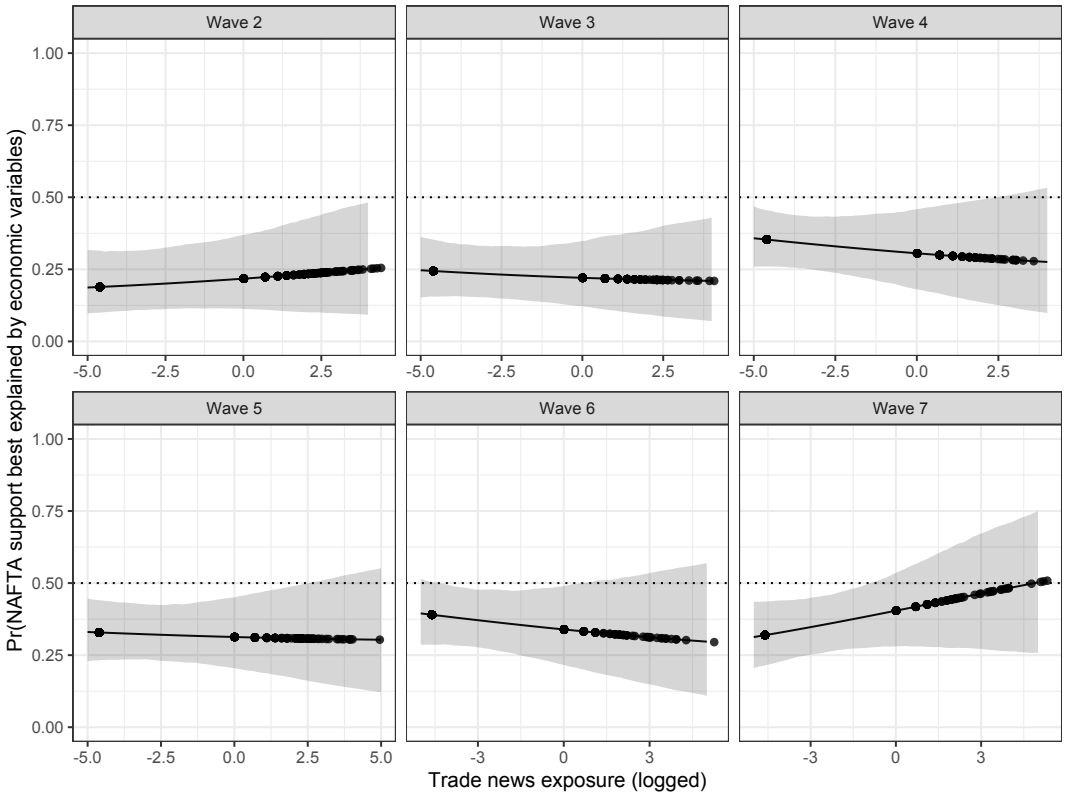
Figure B2 presents the estimated probability of NAFTA attitudes being shaped by economic interests, as a function of respondents' exposure to trade news. The slope of the line in each panel indicates whether the effect of economic interests are magnified or attenuated as respondents increase in exposure to trade news. On the whole, the plot offers little evidence that respondents who consume more news about trade are more likely to anchor their trade preferences on economic interests: apart from wave 7, the slope of the predicted values are flat or negative. Table B3 shows that the superior explanatory power of ideational variables in these models is not merely due to partisanship; when we re-estimate the models, but omit partisanship from the vector of ideational variables, between 64-82% of the observations (depending on the wave) are better explained by ideational variables than by economic interest variables. In Appendix §B.4 we replicate the above analyses, but including community-level measures of economic interests in addition to individual-level measures (Frieden, 2018). We find once again that trade attitudes are better explained by ideational variables than by economic interest measures, and that exposure to news about trade does not moderate the effects of these community-level characteristics.

Figure B1: Respondents exposed to higher levels of trade news no more likely to express trade preferences consistent with economic interests



Respondents exposed to trade news (in turquoise) no more likely to express trade preferences consistent with their economic interests than respondents who were not exposed to trade news (in red). The analysis here defines exposure to trade news as accessing at least four articles about trade during the preceding survey wave, which would place respondents in the 89th-94th percentile of trade news consumption in our sample, depending on the wave.

Figure B2: Finite mixture model: does exposure to trade news magnify the effect of economic interests?



The results from a series of finite mixture models, in which attitudes towards NAFTA are modeled as a function of either economic interests or ideational variables, with the mixing probability between the two as a function of exposure to trade news. The predicted values for Wave 7 displays a positive slope, suggesting that in our final survey wave, the explanatory power of economic increases as respondents increase in trade news exposure. For all other waves, however, the predicted values display a flat (or, for four out of the five waves, negative) slope. Across the study as a whole, then, there is relatively little evidence that exposure to trade news magnifies the effects of economic interests. See Table B3 for the population proportion estimates for each model.

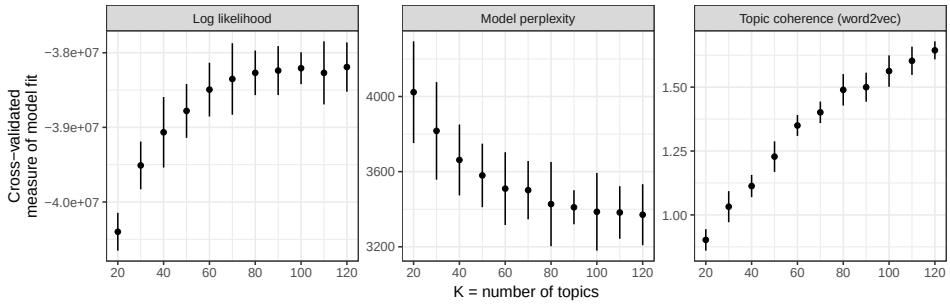
Table B3: Proportion of observations consistent with each set of explanatory variables

	(1)		(2)	
Wave	All Ideational Variables	Economic Variables	Foreign Policy Orientations Only	Economic Variables
2	0.808	0.192	0.817	0.183
3	0.778	0.222	0.787	0.213
4	0.663	0.337	0.695	0.305
5	0.682	0.318	0.701	0.299
6	0.638	0.362	0.636	0.364
7	0.664	0.336	0.698	0.302

Results are π estimates from a set of finite mixture models, denoting the proportion of NAFTA attitiudes in each survey wave consistent with either ideational variables or economic interest variables. Model 2 drops party ID from the model, suggesting the superior explanatory power of ideational variables isn't being driven by partisanship.

B.2 Topic models of trade news

Figure B3: Latent Dirichelet Analysis (LDA) model selection



To characterize the types of articles in which respondents encounter information about trade, the analysis in the main text turn to topic modeling, an unsupervised machine learning technique for the automated content analysis of text (Quinn et al., 2010; Grimmer and Stewart, 2013). We estimate a 100-topic Latent Dirichelet Allocation (LDA) model on the 6894 articles mentioning trade that respondents accessed during the study, based on the model fit statistics presented in Figure B3. 10 of the topics are semantically interpretable and conceptually related to trade. Below, we present a sampling of the articles the LDA indicates is most representative for each of the topics, reproducing the first 2000 characters of each article. The articles show that trade is invoked in a wide number of contexts, sometimes as the main focus of the article (e.g. chickpea farmers suffering as a result of the trade war, the "Trump bump" in car prices as a result of tariffs on imported auto parts, the US and Mexico agreeing to overhaul the North American Free Trade Agreement, the turmoil in the Turkish economy following President Trump's doubling of tariffs on Turkish steel and aluminum, etc.), and other times more indirectly (e.g. the $3 trillion of shipborne trade passing through the South China Sea each year being referenced in the context of military disputes in the region, the trade war being mentioned in the context of greater American disengagement from international cooperation, President Trump claiming that a new trade deal with Mexico will pay for a border wall, Saudi Arabia freezing trade with Canada in response to Canadian criticism of Saudi human rights practices, and so on). In our analysis, we replicate our analyses a number of different ways, from using the most permissive measure of trade news (any article that mentions trade or tariffs), to a more restrictive measure in Figures B4-**??** (any article that mentions trade or tariffs that is also classified by the LDA as pertaining to one of the 10 theoretically-relevant topics), to the most restrictive measure (any article that mentions trade or tariffs that is also classified by the LDA as pertaining to the 5 most topics deemed the most theoretically relevant

based upon the representative documents: agriculture, cars, the economy, NAFTA, and the Turkish financial crisis). Crucially we obtain strikingly similar results regardless of the measure used.

Agriculture

Chickpeas Sit In Silos As Trump's Trade Wars Wage On Chickpeas Sit In Silos As Trump's Trade Wars Wage On Embed Embed Facebook Snow-covered farm fields roll out outside Kendrick, Idaho. Farmers are stumped on what to plant this coming spring, as many of their traditional dryland crops are priced below the cost of production right now. Anna King/Northwest News Network hide caption toggle caption Anna King/Northwest News Network Snow-covered farm fields roll out outside Kendrick, Idaho. Farmers are stumped on what to plant this coming spring, as many of their traditional dryland crops are priced below the cost of production right now. Anna King/Northwest News Network Updated at 9:35 a.m. ET Jan. 3 On a recent rainy day, farmer Allen Druffel stands outside a silo shuffling his feet in the gravel. This co-op bin is where he stores his dried garbanzo beans in the tiny town of Colton, Wash. The place should be busy; trucks should be loading and hauling this year's crop to markets and international ports. But midafternoon, there's just the rain. Since farmers like Druffel brought in this year's crops, hardly any garbanzos – or chickpeas – have moved. "Thirty to 40 percent of our total revenue is in the bin," Druffel says. "And we're not sure what we want to do with it." And it's bad times for lentils and peas, too. In the agriculture industry, these are all called pulse crops. The largest importers of U.S. pulse crops have slapped tariffs on them, and they've been sitting in silos ever since. Enlarge this image Allen Druffel, 34, of Colton, Wash., stands in front of the co-op silos that hold his unsold chickpeas. Last year he was getting 50 cents a pound for his pulse crop. Now, the going price is 18 cents a pound – well below his cost of production. Anna King/Northwest News Network

Cars

August 30, 2018 Get ready for a Trump bump in car prices. As U.S. negotiators close in on a new trade deal with Mexico and Canada, they appear committed to new rules on car production that will require costlier components, which in turn would raise prices. Further provisions would allow for new tariffs on some imported cars that could boost sticker prices by thousands of dollars. The new rules would require more American-made content in cars imported from Canada and Mexico, in exchange for allowing those products to enter the U.S. duty-free. Another measure would raise the portion of a car that must be built by workers earning relatively high wages. Cars coming from Mexico and Canada that don't meet those requirements will be subject to a 2.5% tariff, while cars imported from other countries could face tariffs as high as 25%. The protectionist measures are meant to safeguard U.S. manufacturing jobs. But they come at a cost to consumers. "At the least, the new rules will raise the price of autos for U.S. consumers," writes Chad Bown of the Petersen Institute for International Economics . "The North American auto sector could suffer an even worse blow if the Trump administration imposes new tariffs or quantitative limits on autos and parts not covered by the new deal." Of the 17.3 million cars sold in the United States last year, about 8.6 million were produced in the U.S. That leaves 8.7 million imports. Of the imports, Canada and Mexico each account for about 1.9 million. Popular models made in Mexico include the Ford Fusion, GMC Terrain, Jeep Compass, Honda HR-V and Volkswagen Jetta. Canada produces the Chrysler Pacifica, Ford Edge, Chevy Equinox, Cadillac XTS and Lexus RX450h, among others. The new version of the Toyota RAV4, made in Canada, could cost more on account of President Trump. More The new rules would raise the required amount of North American content in a tariff-free car from 62.5% to 75%. Another rule would require at least 70% of the

Economy

Lucia Mutikani Aug 13th 2018 5:09PM WASHINGTON, Aug 13 (Reuters) - U.S. economic growth will probably accelerate this year before slowing in 2019 to well below the Trump administration's 3 percent target as a fiscal stimulus fades, congressional researchers projected on Monday. In an updated economic outlook, the nonpartisan Congressional Budget Office (CBO) projected that inflation adjusted or real gross domestic product (GDP) would grow 3.1 percent this year, exceeding 2.2 percent growth in 2017 due to lower income taxes, increased government spending and private investment. The government slashed corporate and personal income taxes in January in a $1.5 trillion package and the U.S. Congress passed a $1.3 trillion spending bill in March. This has buoyed consumer and business spending as well as government outlays, which combined with accelerated soybean exports to lift the economy to a 4.1 percent annualized rate in the second quarter from a 2.2 percent pace in the January-March period. The April-June growth rate was the highest in nearly four years. But the CBO said it expected growth to slow in the second half as jolts to consumer spending and agricultural exports either fade or reverse. For instance, some second-quarter soybean exports were aimed at beating Chinese tariffs that took effect in July and cut future shipments. 51 PHOTOS The economy of every US state and DC See Gallery The economy of every US state and DC 51. South Dakota South Dakota was only one of two states with negative GDP growth in Q4 2017, with the state's GDP declining by 0.5%. The state's April 2018 average weekly wage of $754 was fourth-lowest in the country, but its unemployment rate of 3.4% was lower than the average rate of 4.0% among the states and DC. Photo credit: Getty 50. North Dakota North Dakota had the worst year-over-year change in its labor market among the 50 states and DC, with the number of non-farm payroll jobs declining by 1.8% between Apri

NAFTA

U.S., Mexico reach NAFTA deal, turn up pressure on Canada By Roberta Rampton and Jeff Mason Share By Roberta Rampton and Jeff Mason WASHINGTON (Reuters) - The United States and Mexico agreed on Monday to overhaul the North American Free Trade Agreement (NAFTA), putting pressure on Canada to agree to new terms on auto trade and dispute settlement rules to remain part of the three-nation pact. Auto stocks soared and the S&P 500 and the Nasdaq rallied to record highs on the expectation that Canada would sign onto the deal and ease the economic uncertainty caused by U.S. President Donald Trump's repeated threats to ditch the 1994 accord. Details of gains and concessions in the deal were only starting to emerge on Monday. Trump threatened he still could put tariffs on Canadian-made cars if Canada did not join its neighbors and warned he expected concessions on Canada's dairy protections. "I think with Canada, frankly, the easiest we can do is to tariff their cars coming in. It's a tremendous amount of money and it's a very simple negotiation. It could end in one day and we take in a lot of money the following day," Trump said. Negotiations among the three partners, whose mutual trade totals more than $1 trillion annually, have dragged on for more than a year, putting pressure on the Mexican peso <MXN=>and the Canadian dollar <CAD=>. Both currencies gained against the U.S. dollar after Monday's announcement. The political stakes are high for all three countries. Trump and Republicans in the U.S. Congress up for re-election in November want to ensure farmers and other voters whose jobs depend on trade with Canada and Mexico that the deal is sealed. Mexican President Enrique Pena Nieto wants to sign the agreement before leaving office at the end of November, and Canadian Prime Minister Justin Trudeau faces a national election expected by October 2019. Canadian Foreign Minister Chrystia Freeland is expected to travel to Washington for talks on Tuesday.

Turkey

You are using an older browser version. Please use a supported version for the best MSN experience. If Turkey implodes, here's who else will get hurt CNBC 8/10/2018 Silvia Amaro President Trump doubles steel and aluminum tariffs on Turkey as lira plunges CNBC Next Media Animation via Reuters0:57 UP NEXT With financial turmoil in Turkey roiling the country's assets, analysts are busy contemplating which other countries could be exposed and what this could mean for international investors. Quotes in the article -2.99% Turkey has been in economic difficulties since the start of the year. Its currency has lost about 33 percent of its value on the back of large fiscal stimulus, growing inflation and a hefty current account deficit. There's also been intervention from President Recep Tayyip Erdogan on the central bank's policy decisions which continues to surprise investors and accentuate the fall in the currency. European banks Media reports suggested Friday that officials at the European Central Bank are concerned over southern European banks, which have lent significant amounts of money in Turkey. This suggests that investors holding stock in European banks could be at risk. The ECB declined to comment on the reports. Data from the Bank for International Settlements (BIS) – often called the central bank of central banks – shows that Spanish banks are owed $83.3 billion by Turkish borrowers; French lenders $38.4 billion; and banks in Italy are owed $17 billion. Regulators in Europe are reportedly worried that the weaker currency will lead to defaults in foreign loans. Shares of UniCredit, BNP Paribas and BBVA fell around 3 percent Friday morning, after being cited as the European banks with the highest exposure to Turkey. This dragged the wider market down with the pan-European Euro Stoxx 600 down 0.7 percent by 9:00 a.m. London time and the banking index dropping 1.3 percent. Carsten Hesse, a European economist at Berenberg, said in a resear

China

Royal Navy warship 'confronted by Chinese military' in South China Sea Chris Graham September 6, 2018 HMS Albion reportedly passed by the Paracel Islands in recent days to challenge Beijing's 'excessive claims' - PA More A Royal Navy warship has sailed close to islands claimed by Beijing in the South China Sea, a move denounced by China as a "provocation". In a sign of Britain increasingly flexing its military muscle in the region, HMS Albion last week passed by the Paracel Islands, where it was reportedly confronted by the Chinese military. The Albion, a 22,000 ton amphibious warship carrying a contingent of Royal Marines, was on its way to Ho Chi Minh City, where it docked on Monday after a deployment in and around Japan. Beijing dispatched a frigate and two helicopters to challenge the British vessel, but both sides remained calm during the encounter, a source told Reuters. China said Britain was engaged in "provocation" and that it had lodged a strong complaint. In a statement to Reuters, the Foreign Ministry said the ship had entered Chinese territorial waters around the Paracel Islands on August 31 without permission, and the Chinese navy had warned them to leave. "The relevant actions by the British ship violated Chinese law and relevant international law, and infringed on China's sovereignty. China strongly opposes this and has lodged stern representations with the British side to express strong dissatisfaction," the ministry added. "China strongly urges the British side to immediately stop such provocative actions, to avoid harming the broader picture of bilateral relations and regional peace and stability," it said. "China will continue to take all necessary measures to defend its sovereignty and security." Nick Childs of the International Institute for Strategic Studies said the Chinese Navy was becoming more assertive and regularly shadowed vessels in the South China Sea. "No side has much to gain by deliberately spar

Cooperation

Email Bio Follow February 1 The Trump administration announced Friday that the United States will pull out of a nuclear arms control treaty with Russia, ending a cornerstone Cold War agreement and raising fears of a new nuclear arms race in Europe and Asia. President Trump said Russia is violating the 1987 Intermediate-Range Nuclear Forces (INF) Treaty, a charge Moscow denies, leaving the United States at a disadvantage because of its own compliance at a time when global threats have changed considerably in the more than 30 years since the pact was signed. "We cannot be the only country in the world unilaterally

bound by this treaty, or any other," Trump said in a statement. "We will move forward with developing our own military response options and will work with NATO and our other allies and partners to deny Russia any military advantage from its unlawful conduct." [Trump administration gives Russia an ultimatum on Cold War-era arms treaty] The treaty has been a central element of Europe's security strategy for more than three decades and its signing was considered a crucial moment in Cold War arms control, eliminating more than 2,600 missiles and ending a years-long standoff with nuclear missiles in Europe. But the Trump administration has argued that Russia has not been complying with the pact since 2014 and that it puts the United States at a military disadvantage against China, which is not bound by the treaty. The collapse of the pact highlights the cost of poor relations between Washington and Moscow, which even during the Cold War managed to hammer out mutually agreeable arms control pacts. The withdrawal also signals the increasing peril treaties face in a Trump administration that emphasizes national sovereignty over international cooperation. In addition to withdrawing from the INF Treaty, Trump has also left the Iran nuclear deal and Paris climate accord and threatened to pull out of the North Atlantic Treaty Organization and World T

Europe

Email Bio Follow November 11, 2018 PARIS – In the shadow of a grand war memorial here, French President Emmanuel Macron marked the 100th anniversary of the end of World War I by delivering a forceful rebuke against rising nationalism, calling it a "betrayal of patriotism" and warning against "old demons coming back to wreak chaos and death." His words during a solemn Armistice Day ceremony under overcast skies at the foot of the Arc de Triomphe in the heart of the French capital were intended for a global audience. But they also represented a pointed rebuke to President Trump, Russian President Vladimir Putin and others among the more than 60 world leaders in attendance. Speaking in French, Macron emphasized that a global order based on liberal values is worth defending against those who have sought to disrupt that system. The millions of soldiers who died in the Great War fought to defend the "universal values" of France, he said, and to reject the "selfishness of nations only looking after their own interests. Because patriotism is exactly the opposite of nationalism." [As the world marks 100 years since the end of WWI, nationalism is on the rise again] Macron has attempted to stand as a vocal counterweight to Trump, who recently called himself a "nationalist" and has moved to set the United States apart from global treaties, including the Iran nuclear deal, the Paris climate accord and a U.N. program for refugees. Amid growing divisions in Europe that have strained the European Union, Macron defended that institution and the United Nations, declaring that the "spirit of cooperation" has "defended the common good of the world." Morocco?s King Mohammed VI, far left, first lady Melania Trump, President Trump, German Chancellor Angela Merkel and French President Emmanuel Macron attend a commemoration ceremony for Armistice Day. (Benoit Tessier/POOL/EPA-EFE) "By putting our own interests first, with no regard for others, we erase the very thing th

Immigration

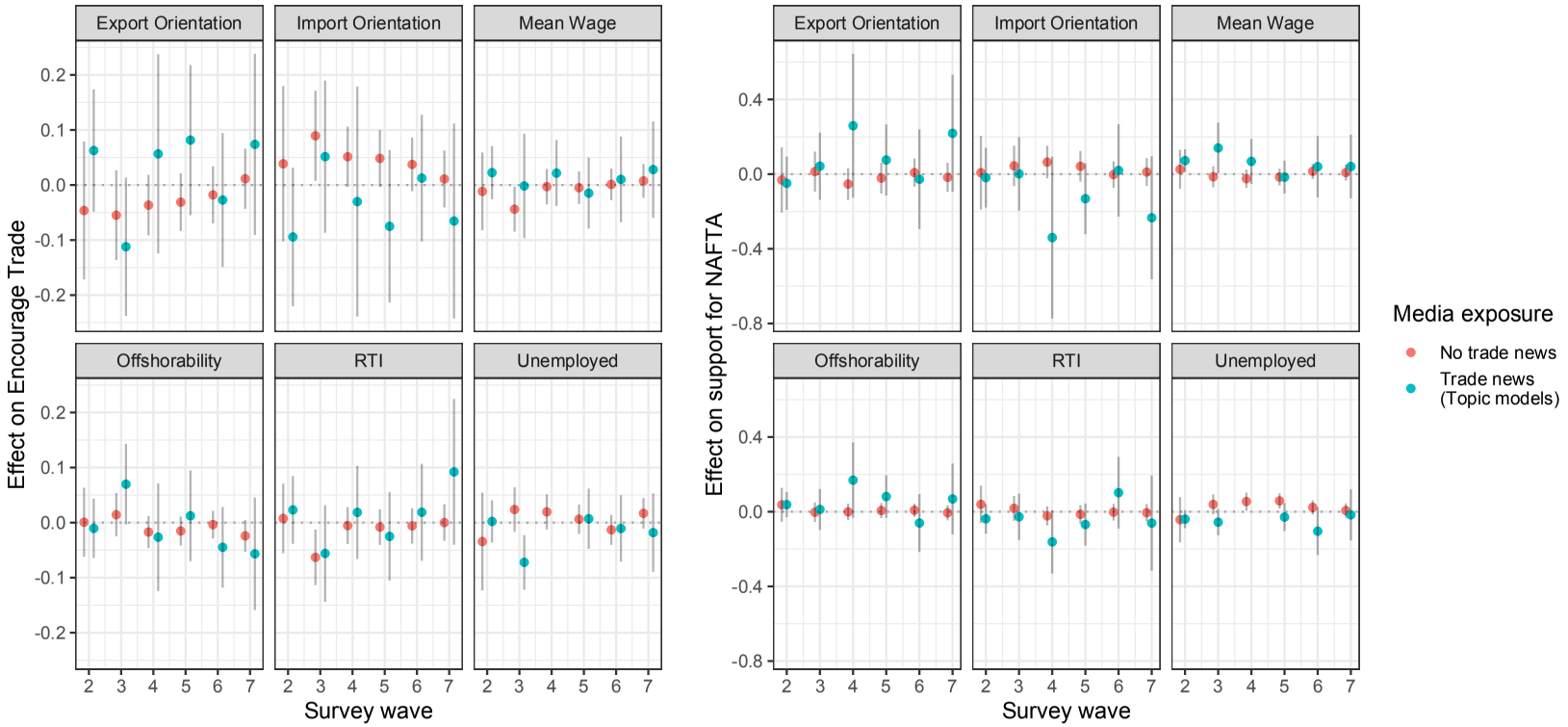
AP FACT CHECK: Trump and the disputed border crisis CALVIN WOODWARD and COLLEEN LONG , January 9, 2019 1 / 5 As seen from a window outside the Oval Office, President Donald Trump gives a prime-time address about border security Tuesday, Jan. 8, 2018, at the White House in Washington. (AP Photo/Carolyn Kaster) WASHINGTON (AP) ? In his prime-time speech to the nation, President Donald Trump declared a border crisis that's in sharp dispute, wrongly accused Democrats of refusing to pay for border security and ignored the reality of how drugs come into the country as he pitched his wall as a solution to varied ills. A look at his Oval Office remarks Tuesday night: DRUGS TRUMP: "Our southern border is a pipeline for vast quantities of illegal drugs, including meth, heroin, cocaine and fentanyl. Every week, 300 of our citizens are killed by heroin alone, 90 percent of which floods across from our southern border." THE FACTS: A wall can't do much about that when drug trafficking is concentrated at land ports of entry, not remote stretches of the border. The Drug Enforcement Administration says "only a small percentage" of heroin seized by U.S. authorities comes across on territory between ports of entry. The same is true of drugs generally. In a 2018 report , the agency said the most common trafficking technique by transnational criminal organizations is to hide drugs in passenger vehicles or tractor-trailers as they drive into the U.S. though entry ports, where they are stopped and subject to inspection. They also employ buses, cargo trains and tunnels, the report says, citing other smuggling methods that also would not be choked off by a border wall. Trump recently denied that traffickers use entry ports at the southern border, contradicting the evidence and assertions of his drug enforcement personnel. Trump stretched credulity even more by comparing the wall money he wants from Congress to the cost of the entire drug problem in the U.S.: "The

Saudi

Saudi Arabia, Canada row over jailed activists ramps up Anuj Chopra with Eric Thomas in Montreal , August 6, 2018 1 / 4 Saudi Crown Prince Mohammed bin Salman has been criticised by rights groups over the targeting of activists and political dissidents Saudi Crown Prince Mohammed bin Salman has been criticised by rights groups over the targeting of activists and political dissidents (AFP Photo/BANDAR AL-JALOUD) Riyadh (AFP) - Saudi Arabia and Canada on Monday showed no signs of backing down in an escalating row over human rights, after Riyadh abruptly cut ties over Ottawa's vigorous calls for the release of activists jailed in the kingdom. The Saudi government expelled Canada's ambassador, giving him 24 hours to leave the country, and recalled its own envoy to Ottawa, while freezing all new trade over what it slammed as "interference" in its internal affairs. Ottawa did not give any ground over the shock expulsion of the envoy, Dennis Horak, with Foreign Minister Chrystia Freeland saying Canada would continue to defend human rights around the world. The rupture, which underscores a newly aggressive foreign policy led by Crown Prince Mohammed bin Salman, comes after Canada denounced a new crackdown

on women and human rights campaigners in the kingdom. "The kingdom announces that it is recalling its ambassador to Canada for consultation. We consider the Canadian ambassador to the kingdom persona non grata and order him to leave within the next 24 hours," the Saudi foreign ministry tweeted early Monday. The ministry also announced "the freezing of all new trade and investment transactions with Canada while retaining its right to take further action". Later Monday, the row appeared to escalate as Riyadh said it will relocate thousands of Saudi students studying in Canada to other countries, and state airline Saudia announced it was suspending flights to and from Toronto. A pro-government Saudi Twitter account provoked outrage when it posted -

Figure B4: Respondents exposed to trade news (as defined by topic model results) no more likely to express trade preferences consistent with economic interests



Respondents exposed to trade news (in turquoise) no more likely to express trade preferences consistent with their economic interests than respondents who were not exposed to trade news (in red). The analysis here defines exposure to trade news as accessing at least one article about trade during the preceding survey wave, consisting of those articles whose highest probability topic in a topic model is one of the ten trade topics presented in section 5.2 of the main text. This threshold would place respondents in the 96th-99th percentile of trade news consumption in our sample, depending on the wave.

Table B4: Test of sociotropic mechanism using topic model results

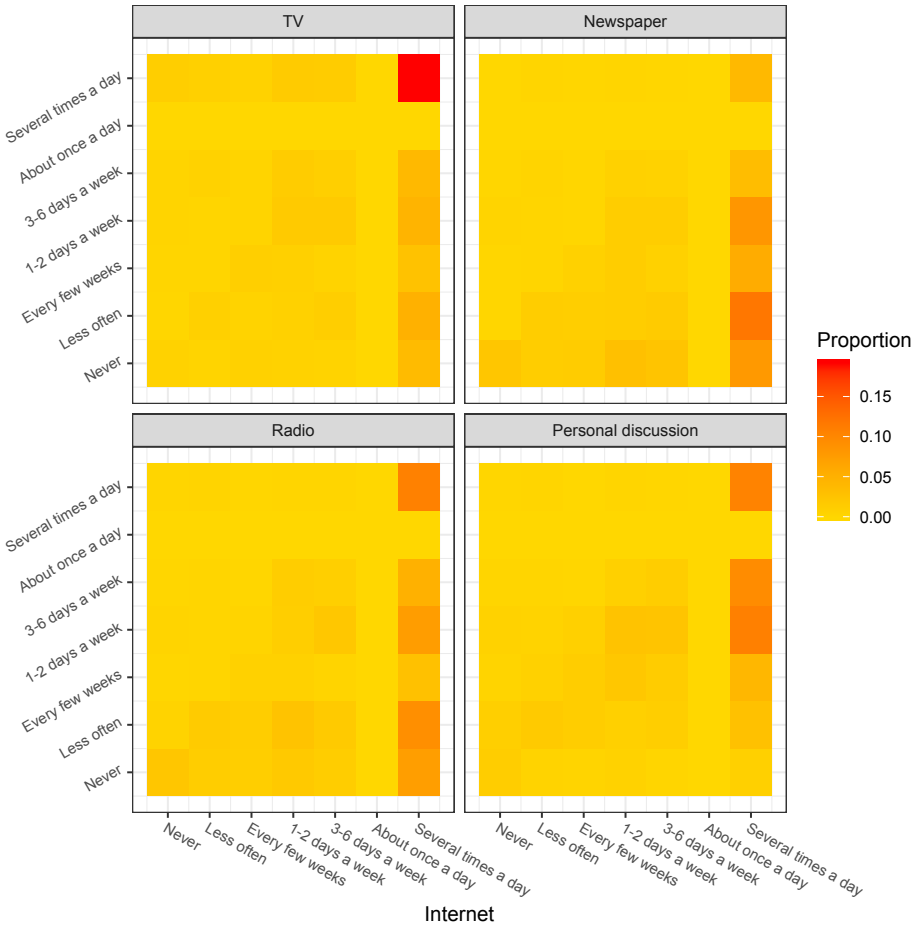
	Encourage Trade				NAFTA			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
No trade news exposure	0.007	0.028**	0.012	0.009	0.012	−0.010	0.006	−0.004
	(0.008)	(0.014)	(0.012)	(0.011)	(0.010)	(0.016)	(0.014)	(0.012)
Positive trade news exposure	0.011	0.053***	0.025*	0.013	−0.004	−0.028	−0.006	−0.021
	(0.009)	(0.018)	(0.015)	(0.014)	(0.011)	(0.021)	(0.018)	(0.016)
Respondent FEs	✓	✓	✓	✓	✓	✓	✓	✓
Trade article measure	All	Topic 1	Topics 1-2	Topics 1-3	All	Topic 1	Topics 1-2	Topics 1-3
N	4,819	6,953	6,953	6,953	4,617	6,665	6,665	6,665
R^2	0.0004	0.002	0.001	0.0002	0.001	0.0003	0.0001	0.0004

*p < .1; **p < .05; ***p < .01. Reference category: Negative trade news exposure

B.3 Comparing types of news media consumption

One of the assumptions driving our empirical strategy in the paper is that individuals who aren't exposed to trade news online are also less likely to be exposed to trade news through other means. We obtain evidence in support of this assumption in Figure B5, which shows that respondents who did not report consuming news online are also less likely to be exposed to trade news through other channels.

Figure B5: Respondents who consume political news through other channels are more likely to access it online as well



This heatmap compares respondents' self-reported levels of news consumption online (on the x-axis) with their self-reported levels of news consumption through other media (television, newspaper, radio, and personal discussion, each depicted on the y axis). The plot shows that in general, the more frequently respondents indicate they access news on the internet, the more frequently they indicate they access news through other media as well. The implication is that it is unlikely that respondents being exposed to news about trade offline are not also being exposed to it online.

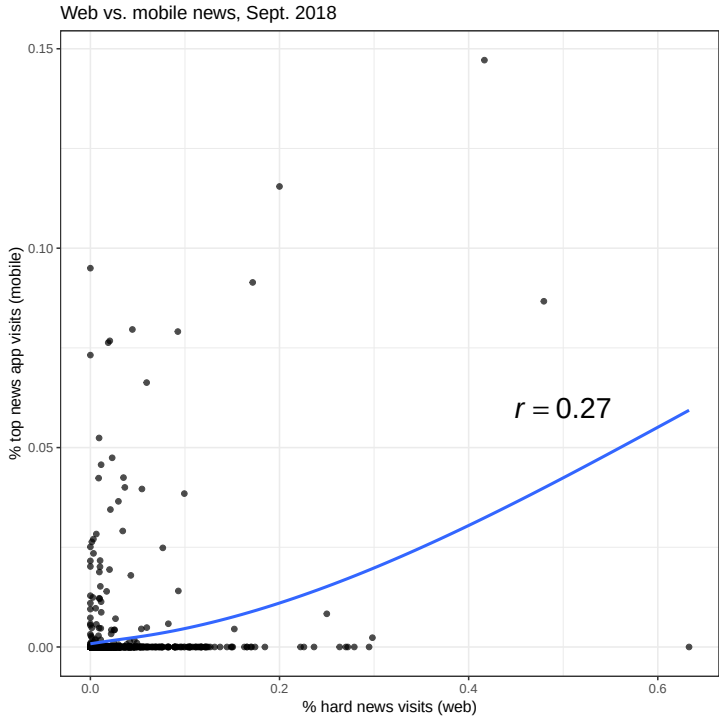
According to the Pew Research Center,[4] 57% of Americans in 2019 said they "often" get news from a mobile device, compared to 30% from desktops or laptops. While our web data cover visits to URLs from

[4] See https://www.pewresearch.org/fact-tank/2019/11/19/americans-favor-mobile-devices-over-desktops-and-laptops-for-getting-ne

desktops and laptops as well as mobile devices, we are more limited in what we observe from respondents' use of news apps. Mobile data logging through news apps is limited to capturing app opens and duration of use, rather than provide the more granular URL-level information we need to capture the content of the news articles. To address this concern, we compare news-related web visits and mobile app use among the same respondents. As Figure B6 shows, the correlation between a measure of the share of web visits to hard news websites and the proportion of news app use is $r = 0.27$.

This calculation is based on respondents for whom we have both web visit and app use data from the month of September 2018 (N=1167). The web visit share is computed using a list of domains classified via a supervised learning model as "hard news" by Bakshy et al. (2015). The news app share is based on the proportion of duration of all app visits from a list of the most-used news-related apps.[5] Figure B6 shows a weak but positive correlation, although there are some respondents for whom we only record news visits in apps, and others for whom we only record it on the web, on average, the more time an individual spends accessing hard news sites in a web browser, the more time they spend consuming it in news apps as well.

Figure B6: Comparing web versus news app visits



[5]The apps are: Apple News, SmartNews, CNN, Flipboard, News Break, feedly news reader, Briefing, WHIO, News Republic - Breaking news, NYTimes - Latest News, USA TODAY, Fox News, Huffington Post, KCRA 3 News and Weather, Daily Mail Online, MetroZone, and BBC News.

B.4 Community-level economic interests

The analysis in the main manuscript focuses on the effects of individual-level economic interests as measures of material interest, rather than community-level ones. It does so partially for reasons of space constraints, but also for two theoretical reasons.

First, the puzzle the manuscript is seeking to resolve in the existing literature is about individual-level interests, rather than community-level ones. Existing scholarship has argued that the reason why individual-level economic interests poorly predict trade attitudes has to do with the role of information (Guisinger, 2017; Rho and Tomz, 2017; Schaffer and Spilker, 2019; Jamal and Milner, 2019; Bearce and Moya, 2020). Our use of individual-level measures of exposure to trade news gives us unique leverage on this question, letting us directly test in a naturalistic setting whether individuals exposed to news about trade are better able to connect their economic self-interest to their trade preferences.

Second, although a promising literature has emerged pointing to the effects of community-level economic interests on political attitudes and behaviors (e.g. Autor, Dorn and Hanson, 2013; Colantone and Stanig, 2018; Ballard-Rosa et al., 2021), it is less directly concerned with the role of information, the focus of this piece, and thus does not provide strong theoretical guidance about how community-level economic factors interact with news about trade in shaping trade preferences. For example, should we envision a positive interaction effect, in which news consumption helps people connect what's going on their local communities to trade policy? Or, should we envision a negative interaction effect, in which individuals in areas strongly affected by trade don't need the news in order to form trade preferences? This literature is thus perhaps better understood as concerned with *how community-level economic indicators shape trade attitudes*, rather than explaining why individual-level economic interests do not, or theorizing about the role of information in the process.

That said, in addition to generating individual-level measures of respondents' economic interests, we also collected a number of community-level measures of economic interests. In order to be consistent with the data used to generate individual-level economic interests, all community-level data are from 2017 unless otherwise noted. The first set of variables are estimated by ZIP Code Tabulation Area (ZCTA), which is used by the U.S. Census Bureau in place of U.S. Postal Service ZIP Codes which are not geographic units.[6] The ZCTA-level data are matched to respondents' self-reported ZIP Codes using a ZIP Code-to-ZCTA crosswalk from the Uniform Data System of the Health Resources and Services Administration (UDS Mapper, 2020).

[6]For a more detailed explanation of the distinction between ZIP Codes and ZCTAs, see https://www.census.gov/programs-surveys/geography/guidance/geo-areas/zctas.html

- Non-Hispanic White share of population: Autor et al. (2020) find that racial makeup is an important moderating variable in determining how trade shocks affect voting patterns at the congressional district level. We use the U.S. Census Bureau's American Community Survey (ACS) five-year estimates of Non-Hispanic White share of population at the ZCTA level (U.S. Census Bureau, N.d.*b*).

- Unemployment rate, labor force participation rate, poverty rate, and disabled rate: Mansfield, Mutz and Brackbill (2019) argue that generalized economic anxiety can elicit anti-trade attitudes even when it is not attributable to trade. We operationalize generalized economic anxiety using ACS five-year estimates of these four measures at the ZCTA level.

- Offshorable employment share: We estimate the share of offshorable employment within each ZIP Code by taking the binary measure of offshorability by occupation (Owen and Walter, 2017) and then aggregating to ZCTA-level estimates using the 2017 ACS five-year estimates of employment by occupation by ZCTA. Since offshorability is estimated at the six-digit SOC level but employment by occupation by ZCTA is only available at the two- and three-digit levels, we first generate estimates of nationwide offshorable employment share within each two- and three-digit occupation category by averaging the subordinate six-digit measures within each category weighted by the national employment of each six-digit occupation. We then produce ZCTA-level estimates using the ACS estimates of employment by two- and three-digit occupation within each ZCTA. We treat this variable as an additional measure of local employment concentration (Guisinger, 2017).

The next set of variables are calculated at the county level due to data availability, and then matched to ZCTAs before being crosswalked to ZIP Codes according to the above procedure. In cases where one ZCTA spans multiple counties, we take the average value of the variable across all applicable counties, weighted by the proportion of the ZCTA's population living in each county as of the 2010 Decennial Census (U.S. Census Bureau, N.d.*a*).

- Intergenerational mobility, 2014: We use Chetty et al. (2014)'s county-level measure of "absolute upward mobility," or the expected income percentile of a child born to parents in the 25th percentile of income, as an additional measure of generalized economic anxiety (Mansfield, Mutz and Brackbill, 2019).

- Chinese import penetration change, 2000–2014: We use Autor, Dorn and Hanson (2019)'s commuting zone-level estimates of the exogenous change in Chinese import penetration from 2000 to

2014. This period covers the surge in Chinese imports following China's entry into the WTO in 2001. Autor et al. (2020) find that localities experiencing a stronger surge in Chinese imports in the decade after China's entry into the WTO tended to elect more politically extreme U.S. House representatives, establishing a link between Chinese imports and political outcomes. Since commuting zones are strict aggregates of counties, the commuting zone-level measures are easily crosswalked to the county level. We isolate the exogenous component of the change in Chinese import penetration using Autor et al.'s instrumental variables method Autor, Dorn and Hanson (2019).

- Manufacturing employment share: Guisinger (2017) finds that trade attitudes of people from localities with a high concentration of employment in certain sectors will tend to reflect the trade-related interests of those sectors. We use the U.S. Census Bureau's County Business Patterns (CBP) data (U.S. Census Bureau, N.d.*c*) to estimate the share of employment in manufacturing industries (NAICS codes 31-33) by county, and then crosswalk these estimates to the ZCTA level.

- Change in manufacturing employment share, 2000–2017: We repeat the same method as above for 2000 CBP data and subtract 2000 manufacturing employment shares from the 2017 estimates. We choose 2000 as the baseline year to match the Chinese import penetration change data from Autor, Dorn and Hanson (2019).

- Employment shares in import-dominant and export-dominant goods industries: Following Guisinger (2017)'s argument about the impact of local sectoral employment concentration on trade attitudes (see under "manufacturing employment share" above), we estimate for each county the shares of employment in import-dominant goods industries and export-dominant goods industries using 2017 CBP data. Import dominance and export dominance by industry is defined the same way as for individual-level economic interests.

Table B5 replicates Table 1 from the main text modeling exposure to trade news, but this time using community-level interests rather than individual-level ones (models 1 and 3), or both community and individual-level economic interests (models 2 and 4); as before, the first half of the models utilize a dichotomous measure of exposure to trade news, while the second half use a negative binomial instead. The dichotomous models find relatively little evidence that community-level conditions predict exposure to trade news, apart from respondents in areas with higher unemployment rates being slightly less likely to consume trade news. In the count models, some of the findings are inconsistent, but we do find evidence that individuals in areas with higher levels of offshorable employment consume a greater number of articles about

Table B5: Exposure to trade news with community-level variables

	Binary (OLS)		Count (negative binomial)	
	(1)	(2)	(3)	(4)
White pop. share	−0.082	−0.148	−0.075	−0.378**
	(0.097)	(0.127)	(0.130)	(0.175)
Unemployment rate	−1.235	−2.300**	−0.266**	−0.341*
	(0.860)	(1.127)	(0.111)	(0.186)
Poverty rate	−0.412	−0.095	−0.169	−0.261
	(0.321)	(0.415)	(0.135)	(0.201)
Disabled rate	0.248	0.776	0.175	0.498**
	(0.536)	(0.700)	(0.145)	(0.196)
Offshorable emp.	−0.296	−0.647	0.228*	0.263*
	(0.486)	(0.621)	(0.121)	(0.154)
Intergenerational mobility	0.003	−0.001	−0.050	−0.110
	(0.005)	(0.006)	(0.104)	(0.127)
Chinese import shock	0.004	−0.017	−0.104	−0.225
	(0.030)	(0.037)	(0.107)	(0.149)
Manuf. emp. share	−0.365*	−0.292	−0.010	−0.201
	(0.200)	(0.253)	(0.162)	(0.193)
Manuf. emp. share change, 2000-17	−0.147	−0.094	−0.046	−0.083
	(0.256)	(0.308)	(0.145)	(0.159)
log(Import-dominant emp.)	−0.011	0.009	0.210	0.301*
	(0.025)	(0.033)	(0.133)	(0.176)
log(Export-dominant emp.)	−0.045	−0.070	−0.470***	−0.425***
	(0.034)	(0.044)	(0.116)	(0.155)
Total URLs visited	0.00001***	0.00001***	2.022***	1.890***
	(0.00000)	(0.00000)	(0.251)	(0.271)
Constant	0.207	0.143	1.210***	0.711***
	(0.303)	(0.446)	(0.096)	(0.128)
Individual-level controls	N	Y	N	Y
N	686	440	686	440
Adjusted R^2	0.139	0.156		
AIC			2631.376	1709.1428

*p < .1; **p < .05; ***p < .01. Explanatory variables standardized for count models

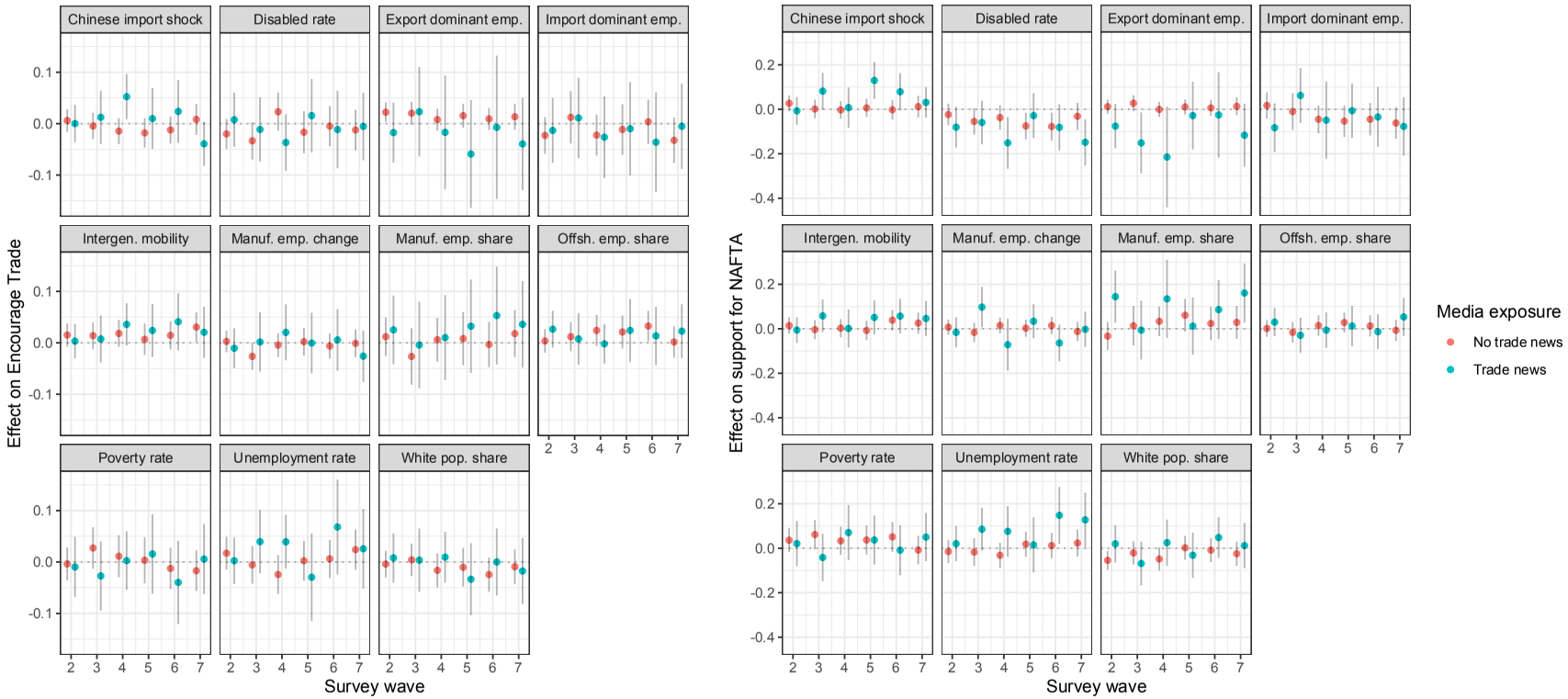
trade, while individuals in areas that feature a higher employment share in export-dominant goods industries consume fewer articles about trade. Generally though, the effects of community economic interests on trade news consumption are weak and inconsistent.

We then proceed to our more central analyses. Figure B7 tests whether community-level economic interests display a larger effect on trade preferences when individuals are exposed to news about trade. For each wave of survey data, we regress trade attitudes on community-level economic variables separately for respondents who saw no trade-related news articles in the preceding survey wave and respondents who saw at least one such article. The plots show that the explanatory power of community-level economic variables are not systematically stronger for respondents who were exposed to trade-related news as compared to those who were not. We obtain similar findings in Figure B8, which plots the marginal effects of community economic interests conditioning on a continuous-level measure of trade news.

Figure B9 uses a finite mixture model to test whether exposure to trade news magnifies the effects of individual- and community-level economic interests more generally. Here, we compare the combined power of individual- and community-level economic variables against that of ideational variables in explaining support for NAFTA. We then model how the relative explanatory power of economic variables varies with exposure to trade-related news; upward-sloping curves would indicate that news exposure magnifies the effects of economic interests. We find a slight upward slope in Wave 2 data and a stronger one in Wave 7, but flat or downward-sloping curves in the other waves. The inconsistent results offer little evidence that trade news exposure increases the influence of economic factors on trade attitudes.

Finally, Table B6 estimates the overall proportion of observations in each survey wave that are consistent with either an ideational model of trade attitudes or an economically-driven model. We conduct the analysis twice, first with both individual- and community-level variables combined in the economic model, and second with only community-level variables in the economic model. As in the main analysis, ideational variables perform better against both individual and community-level economic interest variables combined and community-level variables alone. The economic variables do somewhat better in Wave 7 data, but ideational variables still offer superior explanatory power for a majority of observations in that sample.

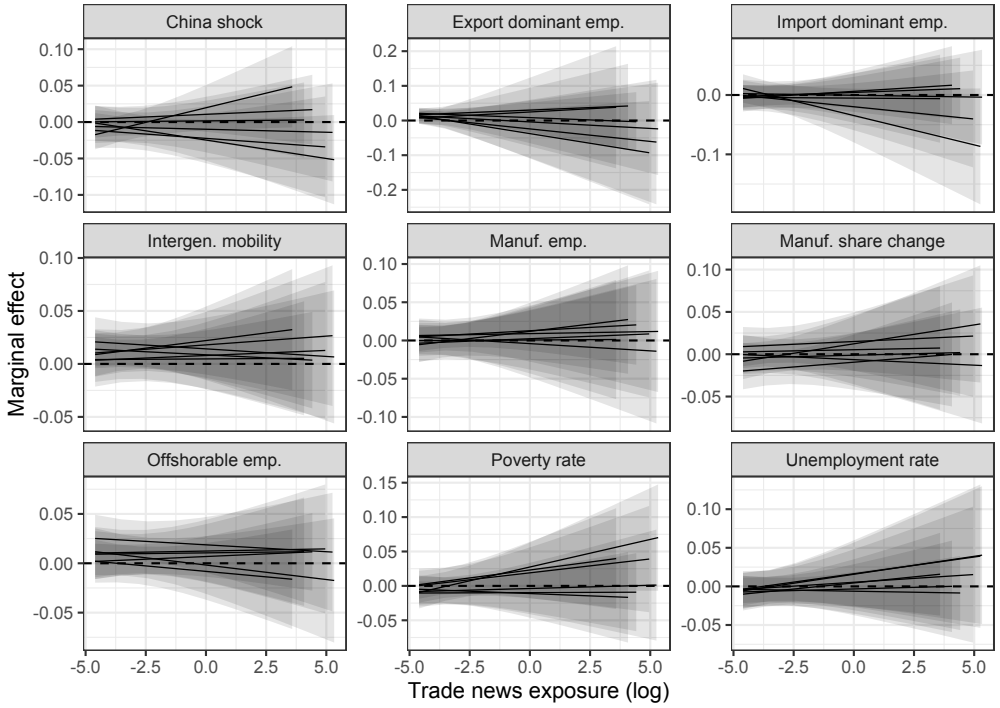
Figure B7: Are community-level economic interests more likely to shape trade preferences when individuals are exposed to news about trade?



Respondents exposed to trade news (in turquoise) are no more likely to express trade preferences consistent with their community-level economic interests than respondents who were not exposed to trade news (in red). The analysis here defines exposure to trade news as accessing at least one article about trade during the preceding survey wave.

Figure B8: Marginal effects of community economic interests do not significantly increase with trade news exposure

(a) Encourage Trade



(b) NAFTA

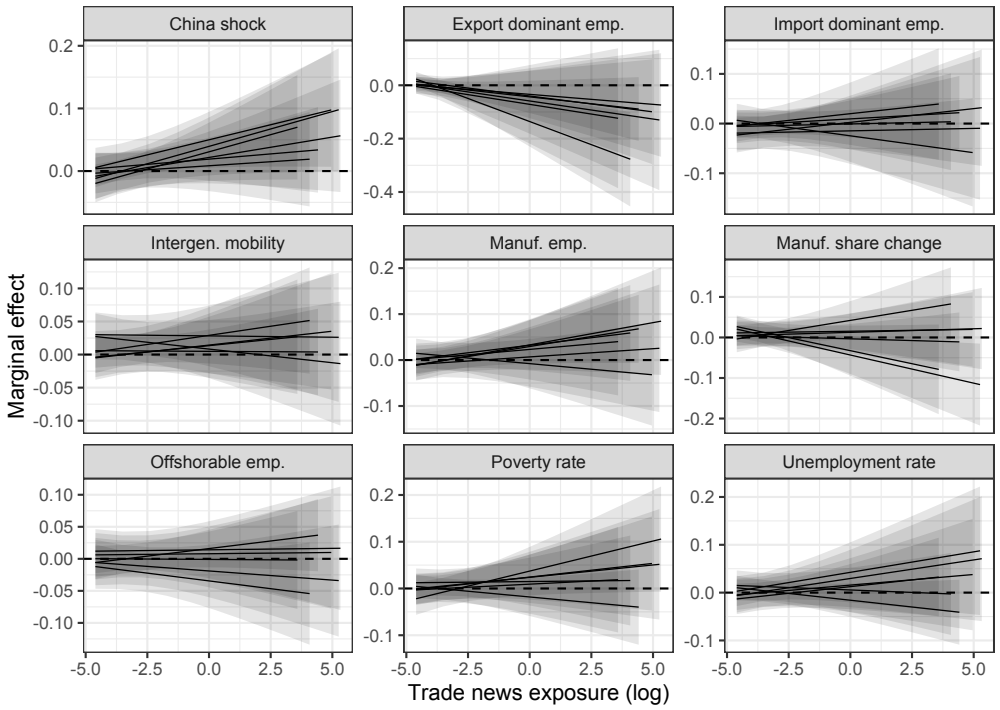
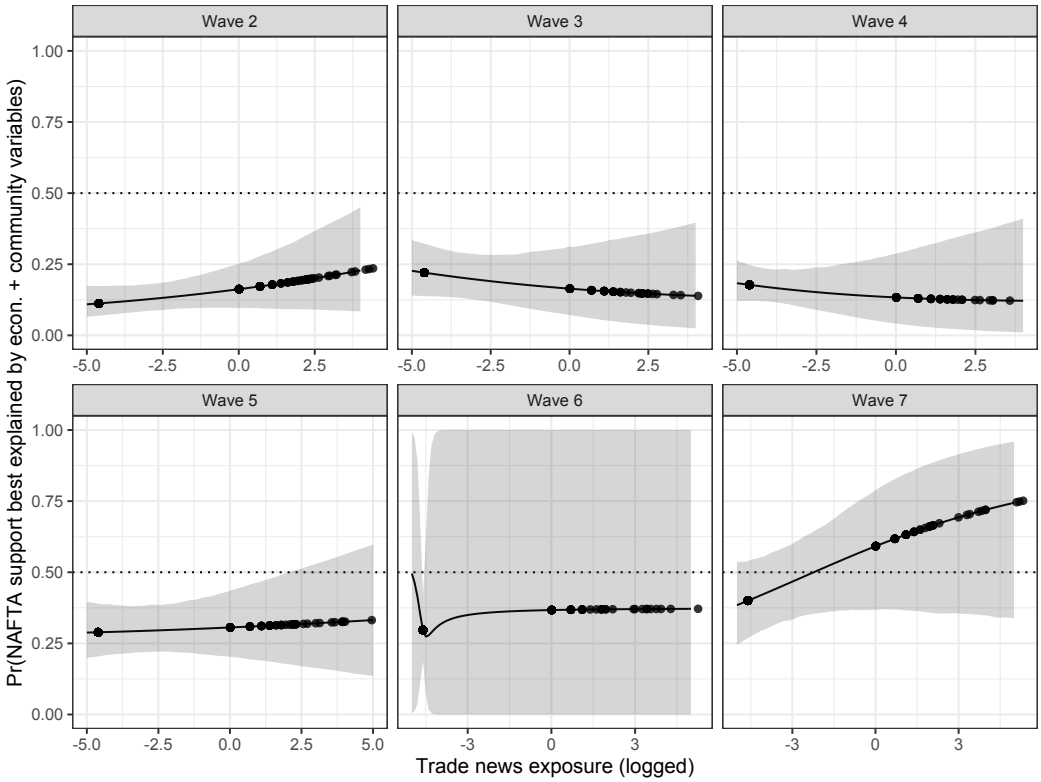


Figure B9: Does exposure to trade news magnify the effect of individual and community-level economic interests?



The results from a series of finite mixture models, in which attitudes towards NAFTA are modeled as a function of either economic interests (both individual-level and community-level) or ideational variables, with the mixing probability between the two as a function of exposure to trade news. The predicted values for Waves 2 and 7 display a positive slope, suggesting that the explanatory power of economic increases as respondents increase in trade news exposure in these two survey waves. For all other waves, however, the predicted values display a flat or negative slope. Across the study as a whole, then, there is relatively little evidence that exposure to trade news magnifies the effects of economic interests. See Table B6 for the population proportion estimates for each model.

Table B6: Proportion of observations consistent with each set of explanatory variables, NAFTA

	(1)		(2)	
Wave	All Ideational Variables	Indiv + Comm. Econ Variables	All Ideational Variables	Community Econ Variables Only
2	0.873	0.127	0.763	0.237
3	0.803	0.197	0.684	0.316
4	0.836	0.164	0.679	0.321
5	0.710	0.290	0.622	0.378
6	0.768	0.232	0.721	0.279
7	0.552	0.448	0.608	0.392

B.5 Supervised machine learning results

For our analysis in the main text, we identify trade news articles using a pure dictionary-based approach; Appendix §B.2 presents robustness checks using topic models to further refine our measure of trade news exposure. As an alternative approach, we also identify articles related to international trade by building a supervised machine learning classifier trained on news articles published during the same period, and which contained topic tags assigned by the news outlets themselves. The goal of this approach is to automatically identify words that are associated with our construct of interest (in this case, that the news article indeed provides information about international trade) based on a sample of manually-labeled articles, and then extrapolate to the entire corpus of news articles that our respondents visited.

In particular, we use a set of 20,366 articles that contain the word "trade" and that were published on the websites for ABC News, Bloomberg News, Breitbart, Buzzfeed, CNBC, DailyKos, HuffingtonPost, The New York Times, Politico, The Hill, USA Today, The Washington Post, The Wall Street Journal or Yahoo News between August 1st, 2018 and March 30th, 2019. These outlets were selected based on both circulation and on the availability of comma-separated article tags. For example, Fox News was excluded because it does not offer article tags.

To select articles that would serve as positive labels in our model, we identified a set of relevant tags that captured our construct of interest. The full list is: "International trade", "Trade regulation", "Global trade", "Sanctions and embargoes", "Trade Policies", "Trade barriers", "Trade agreements", "Trade war", "Tariffs", "NAFTA", "North American Free Trade Agreement", "Embargoes Sanctions", "Trans-Pacific Partnership", "Bilateral Trade Agreement", "Trade Agreement", and "USMCA". We also included the tag "Trade," but with a comma, to exclude tags such as Trader Joe's or World Trade Center. In total, we found that 8.8% of articles with the word "trade" had a topic tag that matched our selection.

Using the text of these articles and the labels, we trained a tree-based classifier called `xgboost` (Chen and Guestrin, 2016), and which has been found to maximize classification accuracy in most tasks (Olson et al., 2017), on a random 80% of our data. With the remaining 20%, which is not used for training, we evaluated how well our model performs out of sample. Our selected model achieves 89% accuracy, 92% precision, and 87% recall. This means that 92% of the articles we predict to be about trade do indeed contain the relevant topic tags; and that we are able to correctly identify 87% of all articles with trade topic tags as such. A cursory look at the words that best predict whether an article is about our construct of interest (e.g. `nafta, tariff, agreement, free, usmca, trans-pacif, north, trump, trade, cliff, mexico`, etc) also confirms the

adequacy of our method.

We also explored alternative methods to pre-process the text (for example, via word embeddings) or to train our model (for example, using a regularized logistic regression), but in our benchmark comparisons we found that our tree-based classifier approach achieved the best performance.

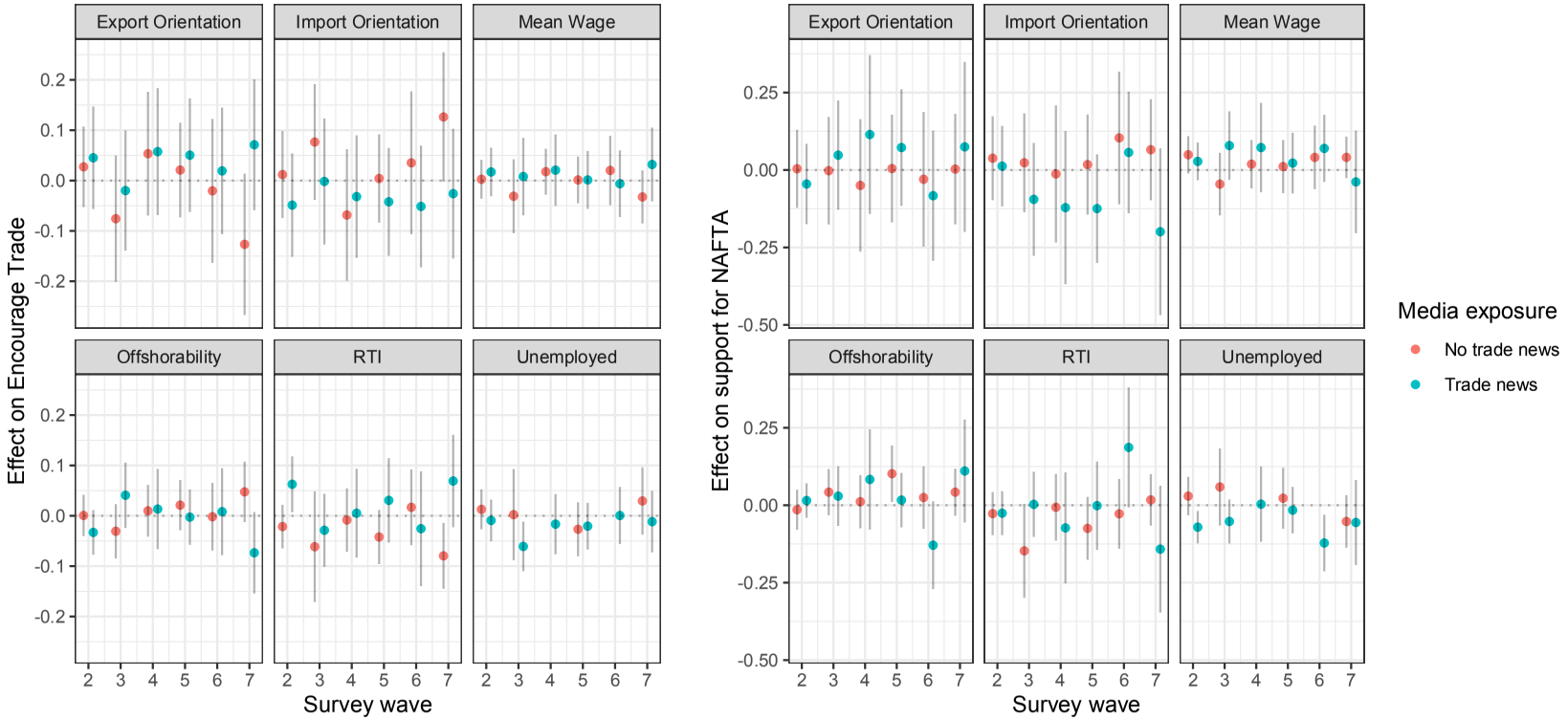
Crucially, as the results on the following pages show, our classifier results are strikingly similar to those reported in the main text: individuals who are exposed to trade articles are no better to connect their material interests with their trade preferences, but respondents in our sample became more supportive of trade the more positive stories about trade they encountered.

Table B7: Exposure to trade news (classifier)

	Binary (OLS)			Count (negative binomial)		
	(1)	(2)	(3)	(4)	(5)	(6)
Some college	0.036	0.028	0.028	−0.019	−0.209	−0.213
	(0.036)	(0.046)	(0.046)	(0.314)	(0.440)	(0.442)
College/university	0.017	0.015	0.014	0.128	−0.083	0.009
	(0.032)	(0.041)	(0.042)	(0.277)	(0.399)	(0.400)
Postgraduate	0.061*	0.042	0.033	−0.038	−0.200	−0.047
	(0.037)	(0.049)	(0.050)	(0.310)	(0.472)	(0.473)
Male	0.022	0.037	0.024	0.175	0.475**	0.418*
	(0.023)	(0.028)	(0.028)	(0.181)	(0.242)	(0.247)
Age	−0.173***	−0.198***	−0.200***	−0.385***	−0.493***	−0.492***
	(0.054)	(0.065)	(0.066)	(0.104)	(0.145)	(0.142)
Party ID	−0.094***	−0.101***	−0.098*	−0.995***	−1.235***	−1.301**
	(0.031)	(0.038)	(0.052)	(0.258)	(0.363)	(0.510)
White	0.066**	0.088**	0.086**	0.404	0.312	0.255
	(0.029)	(0.036)	(0.037)	(0.257)	(0.356)	(0.366)
Political interest	0.181***	0.159***	0.145**	2.297***	2.445***	2.092***
	(0.044)	(0.053)	(0.057)	(0.419)	(0.553)	(0.570)
Unemployed		0.007	0.013		0.094	0.084
		(0.075)	(0.075)		(0.131)	(0.130)
log(Export Orientation)		0.017	0.016		−0.101	−0.070
		(0.022)	(0.022)		(0.301)	(0.298)
log(Import Orientation)		−0.028	−0.024		−0.124	−0.168
		(0.022)	(0.022)		(0.316)	(0.312)
Mean Wage		0.00000	0.00000		0.178	0.176
		(0.00000)	(0.00000)		(0.149)	(0.152)
Offshorability		0.001	0.0004		0.021	−0.020
		(0.001)	(0.001)		(0.136)	(0.138)
RTI		−0.020	−0.013		0.035	0.101
		(0.025)	(0.025)		(0.168)	(0.175)
Isolationism			−0.148***			−0.210
			(0.055)			(0.146)
MI			−0.034			−0.342*
			(0.062)			(0.176)
CI			−0.072			−0.320
			(0.066)			(0.199)
Nat. Attach			−0.099			0.071
			(0.067)			(0.135)
Nat. Chauv			0.082			0.119
			(0.077)			(0.165)
URLs visited	0.00001***	0.00001***	0.00001***	1.841***	1.967***	1.905***
	(0.00000)	(0.00000)	(0.00000)	(0.202)	(0.269)	(0.264)
Intercept	0.029	−0.079	0.130	−2.686***	−2.720***	−2.406***
	(0.060)	(0.102)	(0.141)	(0.484)	(0.701)	(0.724)
N	1,214	818	804	1,214	818	804
R^2	0.131	0.143	0.155			
Adjusted R^2	0.124	0.127	0.133			
AIC				2,258.600	1,508.188	1,497.703

*p < .1; **p < .05; ***p < .01

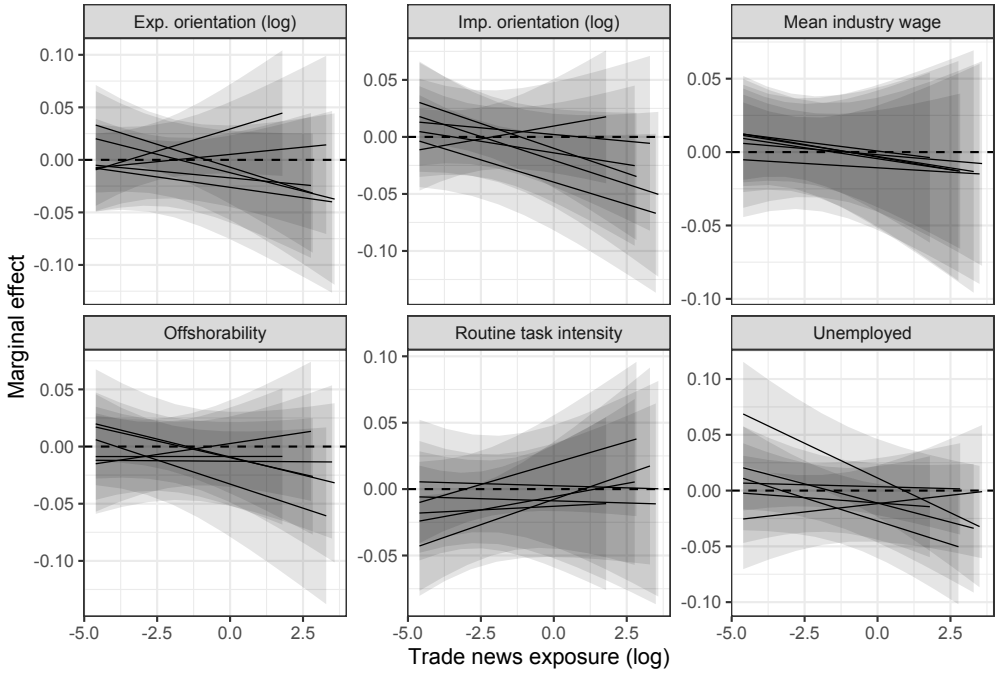
Figure B10: Are economic interests more likely to shape trade preferences when individuals are exposed to news about trade? (Classifier)



Respondents exposed to trade news (in turquoise) are no more likely to express trade preferences consistent with their economic interests than respondents who were not exposed to trade news (in red). The analysis here uses the classifier-based approach to identify trade articles rather than the pure dictionary-based approach from the main text.

Figure B11: Marginal effects of economic interests do not significantly increase with trade news exposure (Classifier)

(a) Encourage Trade



(b) NAFTA

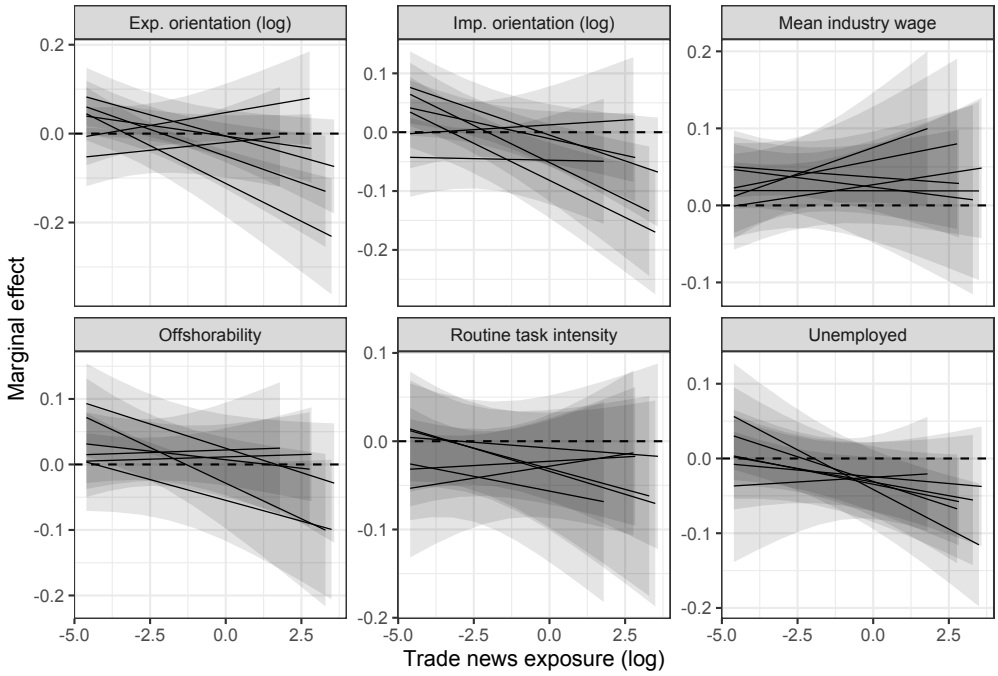


Table B8: Does exposure to news about trade affect trade preferences? (Classifier)

	Encourage Trade		NAFTA	
	(1)	(2)	(3)	(4)
Trade news exposure (logged)	0.00001	−0.003	−0.002	−0.006*
	(0.001)	(0.003)	(0.002)	(0.003)
Trade news sentiment		0.008		0.030
		(0.023)		(0.028)
Trade news exposure x sentiment		0.021*		0.012
		(0.011)		(0.013)
Respondent FEs	✓	✓	✓	✓
N	4,819	4,819	4,617	4,617
R^2	0.00000	0.001	0.001	0.001

*p < .1; **p < .05; ***p < .01

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